CS555

Foundations of Machine Learning

Tahereh Javaheri

This class is ideal for:

- Students interested in learning the foundations of statistics and machine learning
- Students curious about how data can be used to make informed decisions
- Students looking to expand their analytical skills for future career opportunities

Prerequisites:

- No prior experience in R or Python required – but need to be build these skills throughout the semester
- No advanced statistics background needed – we'll start with the basics and progress together
- A willingness to learn and engage with new concepts is essential
- Basic math skills (high school level algebra) will be helpful

By the end of this course, you'll:

- Develop a solid foundation in statistical concepts and machine learning principles
- Learn how to approach real-world problems using data-driven methods
- Be prepared for more advanced courses in data science and AI

This class is for learning – come ready to explore, ask your questions, and discover!

Course Policy

The course is 80% theory and 20% coding.

This course is **not designed to teach R or Python programming**.

The topics in each modulus are subject to change depending on course and students need.

All the material will be posted on Blackboard

Assignments are due before the next class

Office hours: will be on-line and by appointment

TA: Prishita Kohkhar, Prishita@bu.edu, TAs office hours will be announced by TAs

Evaluation 1/3

- **6 assignments, mini quizzes (oral presentation of your homeworks), and final exam.**
  - Assignments:
  - Must be submitted on time
  - Late submissions will not be accepted for grade improvement
  - No exceptions for individual grade boosts at semester's end
  - Policy applies equally to all students for fairness

Evaluation 2/3

- **Homework Grading Criteria:**
  - Accurate result interpretation and reporting
  - Ability to explain code and answer questions
  - 5-12 minute presentation of homework and code description (for in-class review)
  - If using AI tools (e.g., ChatGPT):
    - Acceptable, but must understand and explain the code
    - Ability to describe any modifications made
    - Copy/pasting without understanding is considered plagiarism

Evaluation 3/3

Attendance Policy:

- Class attendance not mandatory, but strongly recommended, because:

  attending students typically achieve higher grades

  In-person interaction helpful for potential recommendation letters

- Mandatory assignment: experiment and study 5 statistical software with reporting on it (5 points).

Grading

A	95-100
A-	90-94.99
B+	85-89.99
B	80-84.99
B-	75-79.99
C+	70-74.99
C	65-69.99
F	<65

Modules 1-6

- Build foundation of data analytics, statistics and machine learning
- Learn ideas behind the techniques
- Help to apply them correctly – know how and when to use them

Module 1

Introduction, Data Summarization, Normal Distribution, Central Limit Theorem, Confidence Interval

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Boston University

Module1/ Lecture1

Table of contents

Rationale for R

1. Freely available under the GNU General Public License: R is an open-source software, which means it can be freely downloaded and used without any licensing fees or restrictions.
2. Pre-compiled binary versions are provided for various operating systems: R is available for Windows, macOS, and Linux, making it accessible to students with different computer setups.
3. Easy to install and ready to use in a few minutes: R can be installed and set up quickly, allowing students to start working with it immediately.
4. Frequent updates: The R community is active, and the software is regularly updated with new features and improvements.
5. A few thousand supplemental packages: R has a vast ecosystem of packages and libraries that extend its functionality, covering a wide range of statistical and data analysis tasks.
6. Open-source with a large support community: R has a large and active community of users and developers, making it easy to find help, resources, and solutions to various problems.
7. Many books, blogs, and tutorials: There is a wealth of educational materials available for learning and mastering R, including books, online tutorials, and user-created content.
8. More popular than major statistics packages: R has gained significant popularity in the statistics and data science communities, surpassing traditional proprietary software like SAS, Stata, and SPSS.

Rationale for Python

1. Freely available under the open-source license: Python, like R, is an open-source programming language that can be freely used and distributed.
1. Pre-compiled binary versions are provided for various operating systems: Python is available for Windows, macOS, and Linux, making it accessible to students with different computer setups.
2. Easy to install and ready to use in a few minutes: Python can be installed and set up quickly, allowing students to start working with it immediately.
3. Frequent updates: The Python community is active, and the language is regularly updated with new features and improvements.
4. Extensive library ecosystem: Python has a vast collection of libraries and packages, such as NumPy, SciPy, and Matplotlib, that provide powerful statistical and data analysis capabilities.
5. Open-source with a large support community: Python has a large and active community of users and developers, providing a wealth of resources, tutorials, and support.
6. Increasingly popular in data science and analytics: Python is gaining significant traction in the data science and analytics fields, making it a valuable skill for students to acquire.
7. Complementary to R: While R is primarily focused on statistical computing, Python offers a more general-purpose programming language that can be used for a wider range of tasks, including data manipulation, visualization, and machine learning.

R installation

Windows:

1. Go to the R project website (https://www.r-project.org/).
2. Click on "Download R" and select the link for "Download R for Windows".
3. Choose the latest version of R for Windows and download the installation file.
4. Run the installation file and follow the on-screen instructions to complete the installation.

macOS:

1. Go to the R project website (https://www.r-project.org/).
2. Click on "Download R" and select the link for "Download R for (Mac) OS X".
3. Choose the latest version of R for macOS and download the installation file.
4. Run the installation file and follow the on-screen instructions to complete the installation.

Linux:

1. Open your Linux distribution's package manager (e.g., apt, yum, or dnf).
2. Search for the "r-base" package and install it using the package manager's command-line interface or graphical user interface.

Python installation

Windows:

1. Go to the Python website (https://www.python.org/).
2. Click on the "Downloads" link and select the latest version of Python for Windows.
3. Run the installation file and follow the on-screen instructions to complete the installation.

macOS:

1. Go to the Python website (https://www.python.org/).
2. Click on the "Downloads" link and select the latest version of Python for macOS.
3. Run the installation file and follow the on-screen instructions to complete the installation.

Linux:

1. Open your Linux distribution's package manager (e.g., apt, yum, or dnf).
2. Search for the "python3" package and install it using the package manager's command-line interface or graphical user interface.

Additional Guidance

- For detailed, step-by-step instructions on installing R and Python on different operating systems:
  - R installation guide: https://cran.r-project.org/doc/manuals/r-release/R-admin.html
  - Python installation guide: https://docs.python.org/3/using/index.html

- Once the software is installed, students can explore the integrated development environments (IDEs) or code editors, such as RStudio for R and Jupyter Notebook or Visual Studio Code for Python, to start working with the languages.

Basic R and Python Data Types

 # numeric types: integer, double

> 348

> 348.19

 # Characters

> "my string"

 # logical

> TRUE

> FALSE

 # Arithmetic operators as you'd expect

> Addition: 42 + 1

*> Multiplication: 1 * 2*

> Exponentiation: 2^4

 # Logical Operators:

OR: TRUE | FALSE

Inequality: 1 + 7 != 7

AND: TRUE & FALSE

NOT: !TRUE

Less than: 1 < 7

Greater than: 7 > 1

Less than or equal to: 1 <= 7

Greater than or equal to: 7 >= 1

Equality: 1 == 1

Inequality: 1 != 7

 # numeric types: integer, floats

Integers: `348`

Floats: `42.0`

 # Characters

Strings: `"my string"`

 # logical

True: `True`

False: `False`

 # Arithmetic operators as you'd expect

Addition: `42 + 1`

Multiplication: `1 * 2`

Exponentiation: `2 ** 4`

#Logical Operators:

OR: `True or False`

Inequality: `1 + 7 != 7`

AND: `True and False`

NOT: `not True`

Less than: `1 < 7`

Greater than: `7 > 1`

Less than or equal to: `1 <= 7`

Greater than or equal to: `7 >= 1`

Equality: `1 == 1`

Inequality: `1 != 7`

Basic R and Python Data Types

```
# Variable Assignment:
my_number <- 483

#Type Checking:
#typeof(my_number)
Output: "double"

#Type Conversion:
my_int <- as.integer(my_number)
typeof(my_int)
Output: "integer"

#Vectors:
Create a vector: my_vec <- c(1, 2, 67, -8)
Get vector properties:

length(my_vec)
Output: 4

#Access elements:
my_vec[3]
Output: 67
my_vec[3:4]
Output: 67 -8
```

```
# Variable Assignment:
my_number = 483

#Type Checking:
type(my_number)
Output: <class 'int'>

#Type Conversion:
my_int = int(my_number)
type(my_int)
Output: <class 'int'>

#Vectors:
Create a list: my_list = [1, 2, 67, -8]
Get list properties:
len(my_list)
Output: 4

#Access elements:
my_list[2]
Output: 67
my_list[2:4]
Output: [67, -8]
```

Working directory - Setting/Getting

 # Working Directory:

Get current working directory:

getwd()

Set working directory:

setwd("/YOUR-HOME-FOLDER/YOURFOLDER")

On Windows:

Remember to use double backslashes or

use a single forward slash "/"

 # List all the objects in the current workspace

> *setwd("C:/Users/xyz/Documents/work/R")*

You can use the RStudio menus to set your working directory.

 # Working Directory:

Get current working directory:

import os

os.getcwd()

Set working directory:

os.chdir("/YOUR-HOME-FOLDER/YOURFOLDER")

The key differences are the syntax for variable assignment (`<-` in R vs `=` in Python) and the data structure for vectors (R uses `c()` while Python uses `[ ]`). Otherwise, the core concepts and operations are very similar between the two languages.

Reading Data into R and Python

```r
# Read a Comma-Separated Values (CSV) data file

# First Line is the header, default value for header is TRUE

df <- read.csv("filename")
```

```r
# Explicitly specify header

df <- read.csv("filename", header=TRUE)
```

```python
# Read a Comma-Separated Values (CSV) data file
import pandas as pd

# Read CSV file (first line is header by default)
df = pd.read_csv("filename")
```

```python
# Explicitly specify header
df = pd.read_csv("filename", header=0)
```

R Libraries and Packages

```
# Install a package (only need to do it once)
install.packages("package_name")

# Access the package
library("package_name")

# View a list of installed packages
library()

# End R session
q()

# It will prompt: Save workspace image? [y/n/c]
# y - yes
# n - no
# c - cancel

# Save content of the current workspace into .Rdata file
save.image()
save.image(file = "abc.Rdata")

# Save some objects of the current workspace into the file
save.image(a, b, file = "abc.Rdata")
```

```
# Install a package (using pip from command line, not within Python)
pip install package_name

# Import a package
import package_name

# View a list of installed packages (from command line)
# pip list

# End Python session
exit()


df = pd.DataFrame({'A': [1, 2, 3], 'B': [4, 5, 6]})
# Save to CSV
df.to_csv('mydata.csv', index=False)


# Save to Excel
df.to_excel('mydata.xlsx', index=False)
```

Statistics

Statistics

▹ A science that deals with the **collection, classification, analysis, and interpretation** of data.

▹ Deals with data collection, evaluation and interpretation.

▹ Statisticians use data to find patterns, answer important scientific questions and draw conclusions.

Two main areas of statistics:

▹ **Describing data** (including numerical and graphical summaries)

▹ **Drawing conclusions** about data (making estimates, predictions,

and decisions) from data collected via sampling

Fundamental Elements of Statistics

▹ An **experimental unit** (or observational unit) is an object (for example, a person, thing, or event) about which we collect data about.

▹ A **population** is a set of units (for example, people, objects, or events) that we are interested in studying.

▹ When studying a population, we focus on one or more characteristics of the units of the population. We call these characteristics **variables**.

Fundamental Elements of Statistics

Variables can be classified into one of two general types:

- **Quantitative**

- **Qualitative (also called Categorical):**

Quantitative Variables

Variables can be classified into one of two general types:

1. **Quantitative**

▹ These are numerical and can be measured and (howmany?; how much?; or how often?)
They are further divided into two types:

a. Continuous: Can take any value within a range
Examples: height, weight, number of houses sold

b. Discrete: Can only take specific values (often integers), number of pets in a household

Qualitative Variables

Variables can be classified into one of two general types:

2. Qualitative (also called Categorical):

▷ Place experimental units into categories

▷ Qualitative data are data about categorical variables (what type?)

▷ Examples: hair color, religion, political party

▷ Categorical variables can be ”ordered levels” are called ”ordinal”.
For example quality of a product can be answered with: very unsatisfied, unsatisfied, neutral, satisfied and very satisfied.
a. Nominal: Categories with no inherent order: no natural hierarchy like gender, race, color, material status, blood type, country of origin
b. Ordinal: Categories with a meaningful order

Categorical variables: can be divided into distinct categories or groups.
Ordered levels: refers to categories that have a natural, meaningful order or ranking.
Ordinal variables: categorical variables with ordered levels: satisfaction level, education level (e.g., high school, bachelor's, master's, doctorate)

Qualitative Variables

Variables can be classified into one of two general types:

Qualitative (also called Categorical):

- ▹ Place experimental units into categories, examples:
  - In a study on the effect of diet on health, you might categorize participants (experimental units) into groups like "vegetarian," "omnivore," and "vegan" (categories).
  - In a marketing survey, you might categorize customers (experimental units) by age groups like "18-25," "26-40," "41-60," and "60+" (categories)
- ▹ Answer "what type?" questions

a. Nominal
Distinct categories with no inherent order
Examples: colors, gender, blood types
Analysis: frequencies, percentages, chi-square tests
b. Ordinal
Categories with a meaningful order
Example: product satisfaction (very unsatisfied to very satisfied), education level, Likert scales

Characteristics of categorical variables:
Mutually exclusive categories
Exhaustive (include all possible options)

Data Summarization

Quantitative Data Summaries

Numerical Summaries focus on measures that describe the center and the spread.

▷ Mean

▷ Median

▷ Variance

▷ Standard Deviation

▷ Quartiles

Graphical Summaries

▷ Histograms – one of the most popular graphical summary of quantitative variables; data are first categorized into classes of equal width and then frequencies is calculated.

▷ Box plots - the median, minimum, maximum, 1st and 3rd quartiles are used to create box plots

Qualitative Data Summaries

Numerical Summaries:

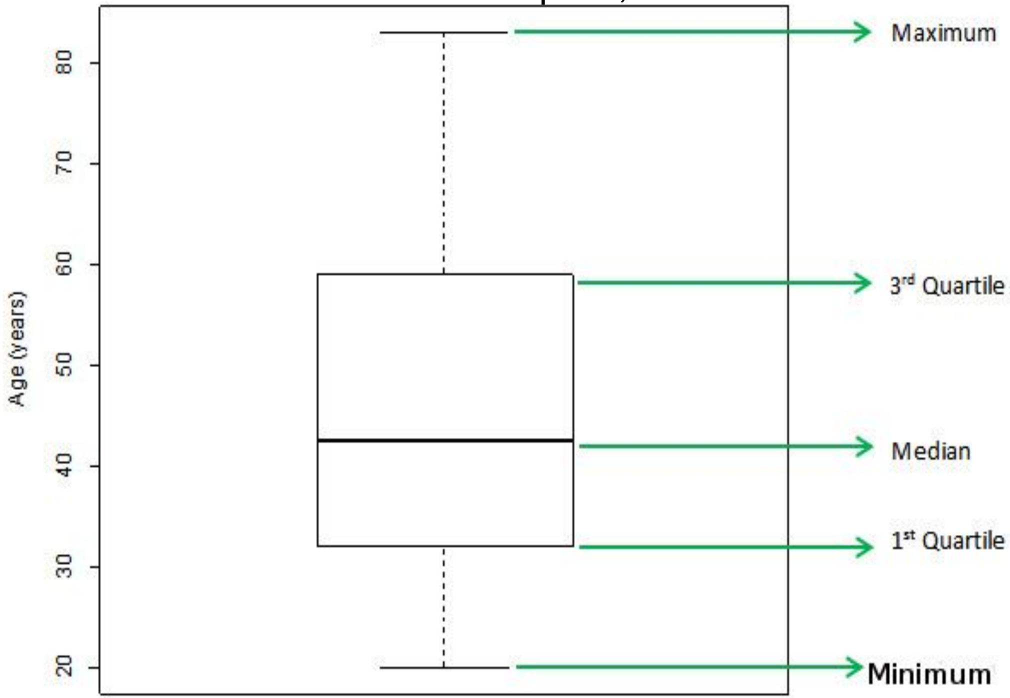
1. Class frequency:
   - Definition: The count of observations in each category; represents the raw count of observations or absolute size in each category
   - Example: If surveying 100 people about their favorite color, 30 might choose blue
2. Class relative frequency:
   - Definition: The proportion of observations in each category relative to the total
   - Calculation: (Class frequency) / (Total number of observations)
   - Example: For blue in the color survey: 30/100 = 0.30 or 30%

Graphical Summaries:

1. Pie Charts:
   - Show the proportion of each category as a slice of a circular "pie"
   - Good for displaying relative sizes of categories
2. Bar Graphs:
   - Use rectangular bars to represent the frequency or relative frequency of each category
   - Useful for comparing categories and showing exact values

Boxplots

box-and-whisker plots,



Display the distribution of a dataset
Show central tendency, spread, and potential outliers
Components:
Box: Represents the interquartile range (IQR)
Bottom of box: First quartile (Q1, 25th percentile)
Line inside box: Median (Q2, 50th percentile)
Top of box: Third quartile (Q3, 75th percentile)
Whiskers: Extend from the box
Outliers: Individual points beyond the whiskers
Key features:
Easily compare distributions across groups
Identify skewness and symmetry
Spot potential outliers
Show the five-number summary (minimum, Q1, median, Q3, maximum)
Advantages:
Compact representation of large datasets and facilitate quick comparisons between datasets
Robust against outliers
Common applications:
Comparing distributions across different groups or categories
Identifying trends in data over time
Quality control in manufacturing processes

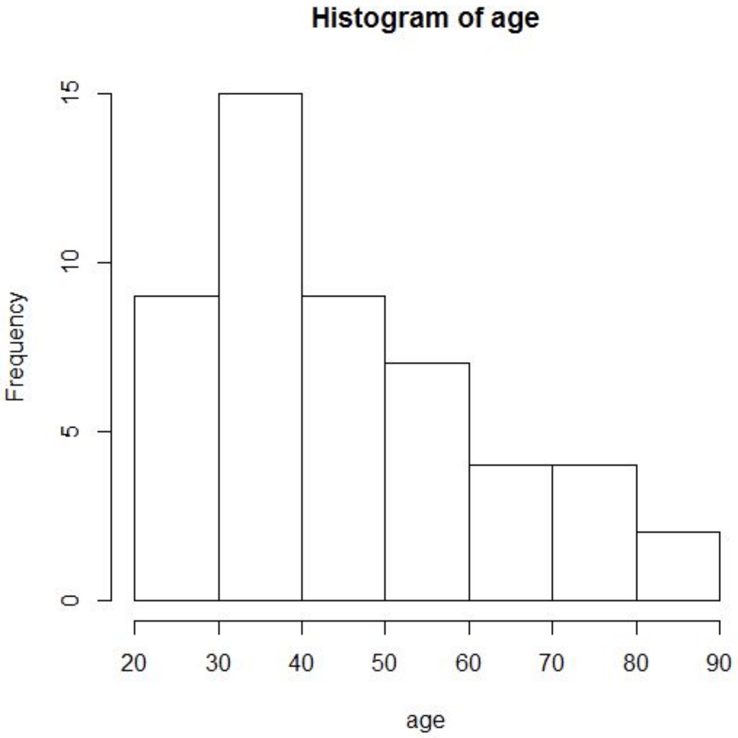
Histograms

Display the distribution of a continuous variable
Show the frequency or density of data points across different intervals
Components:
Vertical axis: Frequency or density of data points
Horizontal axis: Range of the variable, divided into intervals (bins)
Bars: Represent the frequency or density within each interval

Key features:
Shape of the distribution (e.g., normal, skewed, bimodal)
Central tendency
Spread of the data
Presence of gaps or unusual patterns

Construction:
Divide data range into intervals (bins)
Count data points in each interval
Draw bars with heights proportional to counts

Applications:
Analyzing data distributions and identify patterns and outliers in va
Quality control in manufacturing
Examining demographic data

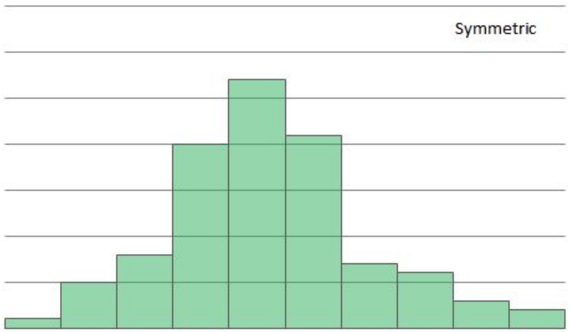


Histograms

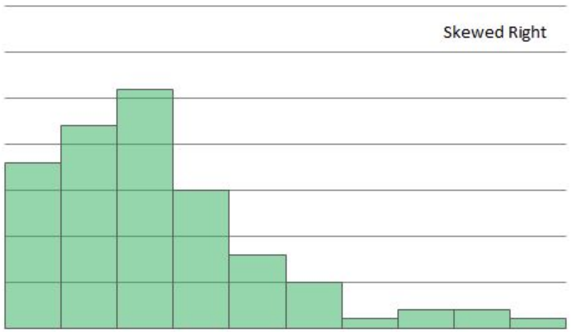
Skewness:
A measure of asymmetry in a distribution
Indicates whether data is concentrated on one side

Right-skewed distribution (positively skewed):
The right tail of the histogram is longer
Most data concentrated on the left
Mean is typically greater than median
Examples: income distributions, home prices

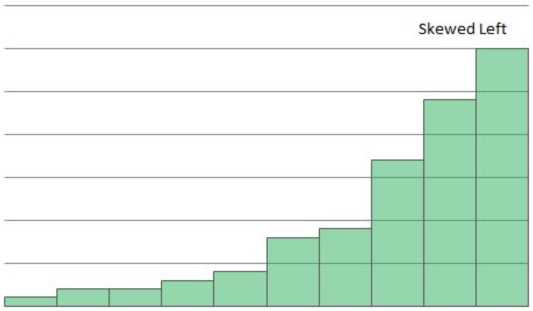
Left-skewed distribution (negatively skewed):
The left tail of the histogram is longer
Most data concentrated on the right
Mean is typically less than median
Examples: age at death, exam scores in an easy test



Symmetric



Skewed right



Skewed left

Quantitative data summaries

▷ The mean and median are numerical measures of the center of a distribution.

▷ The population mean is the mean of all observations for the entire population.

▷ The sample mean for a sample of size n is:

▷ The variance and the standard deviation measure how far each observation is from the mean:

R/Python Functions - Quantitative data summaries commands

```
> mean(data$variable)

> median(data$variable)

> min(data$variable)

> max(data$variable)

> quantile(data$variable)

> var(data$variable)

> sd(data$variable)

> summary(data$variable)
```

```
import pandas as pd
import numpy as np

data = pd.DataFrame({'variable': [1, 2, 3, 4, 5]})
np.mean(data['variable'])

np.median(data['variable'])

np.min(data['variable'])

np.max(data['variable'])

np.quantile(data['variable'], [0.25, 0.5, 0.75]), (for 25th, 50th, and 75th percentiles)

np.var(data['variable'])

np.std(data['variable'])

data['variable'].describe(), provides count, mean, std, min, 25%, 50%, 75%, and max)
```

R/Python Functions - R Functions - Graphical data summaries

Histograms

import matplotlib.pyplot as plt

> hist(data$variable)

plt.hist(data['variable'])

> hist(data$variable, bins)
specify the number of bins

plt.hist(data['variable'], bins=10)
# specify number of bins

> hist(data$variable, breaks=c(x,y,z..))
specify cutpoints

plt.hist(data['variable'], bins=[x, y, z])
# specify cutpoints

> hist(data$variable, breaks=seq(a,b,by=c))
specify cutpoints

plt.hist(data['variable'], bins=np.arange(a, b, c))
# specify cutpoints

Boxplots

> boxplot(data$variable)

plt.boxplot(data['variable'])

General Aesthetic Improvements

Labeling

- Title: main=”Histogram of xyz”
- X-axis label: xlab=”Nile flow”
- Y-axis lable: ylab = ”Frequency”

```
plt.title("Histogram of xyz")
plt.xlabel("Nile flow")
plt.ylabel("Frequency")
```

Colors

http://www.stat.columbia.edu/~tzheng/files/Rcolor.pdf

Controlling the window

- X-axis: xlim=c(min, max)
- Y-axis: ylim=c(min, max)

```
# Use color names or hex codes
plt.xlim(min, max)
plt.ylim(min, max)
```

Combining plots

```
par(mfrow=c(2,2)) # 2 by 2 panels
par(mfrow=c(1,1)) # Go back to single graph mode

library(ggplot2)
library(patchwork)
P1 <- ggplot(data, aes(x = variable)) + geom_histogram()
P2 <- ggplot(data, aes(y = variable)) + geom_boxplot()

P1 + P2  # This will display the plots side by side
```

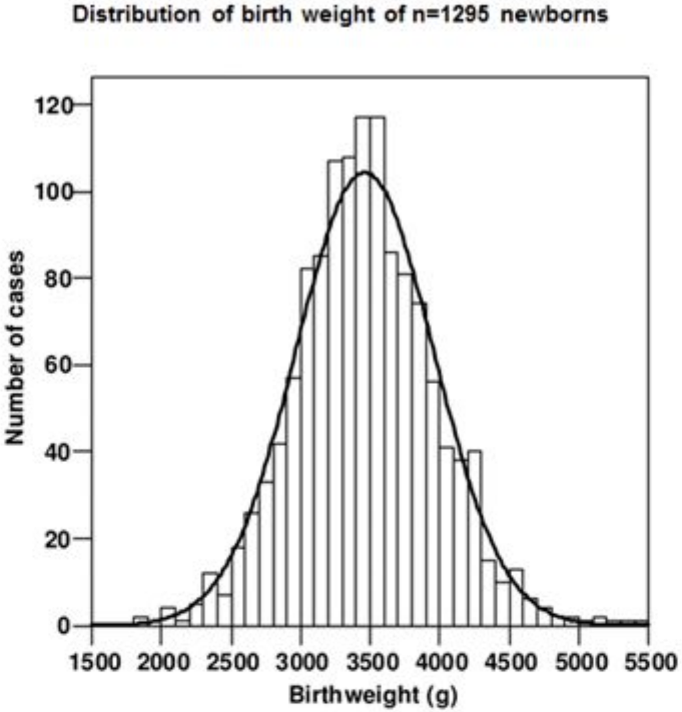
```
fig, axs = plt.subplots(2, 2)
plt.close()  # Go back to single graph mode
```

Normal Distribution

Normal Distribution - Example Curve

A 2006 paper (Low Birth Weight, a Risk Factor for Cardiovascular Diseases in Later Life, Is Already Associated with Elevated Fetal Glycosylated Hemoglobin at Birth) showed a histogram of the birth weights of newborns in their sample.

Q-Q plot, Shapiro-Wilk test

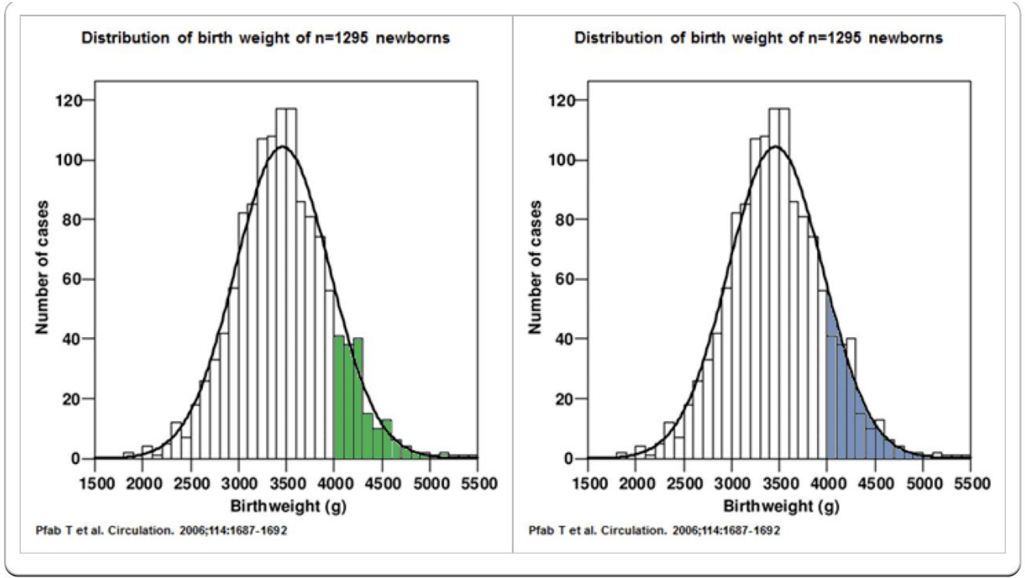


Pfab T et al. Circulation. 2006;114:1687-1692

The proportion of newborns who weighed > 4000g

Using the histogram, it is the proportion of the area of the bars of the histogram to the right of 4000g (where the area of the bars shaded in green).

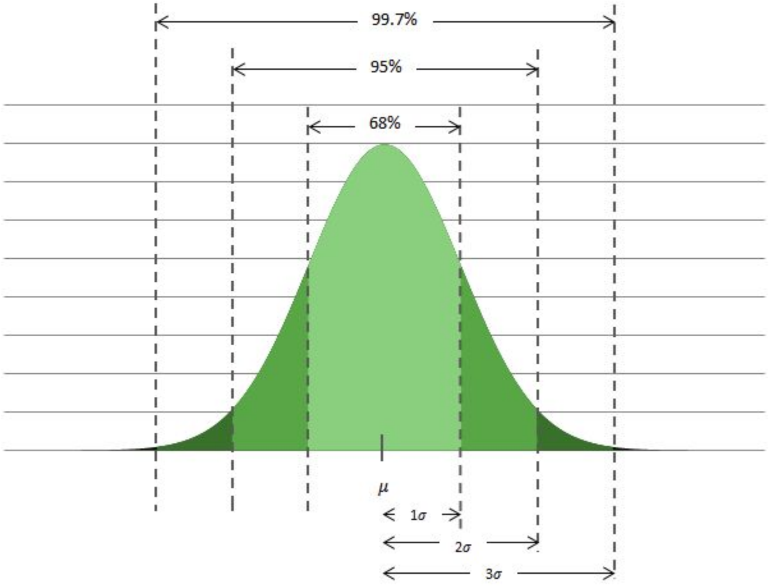
Alternatively, if we scale the curve such that the total area under the curve is 100%, then the area to the right of 4000g under the curve (shaded in blue) would be an approximation of the proportion of infants weighing more than 4000g.



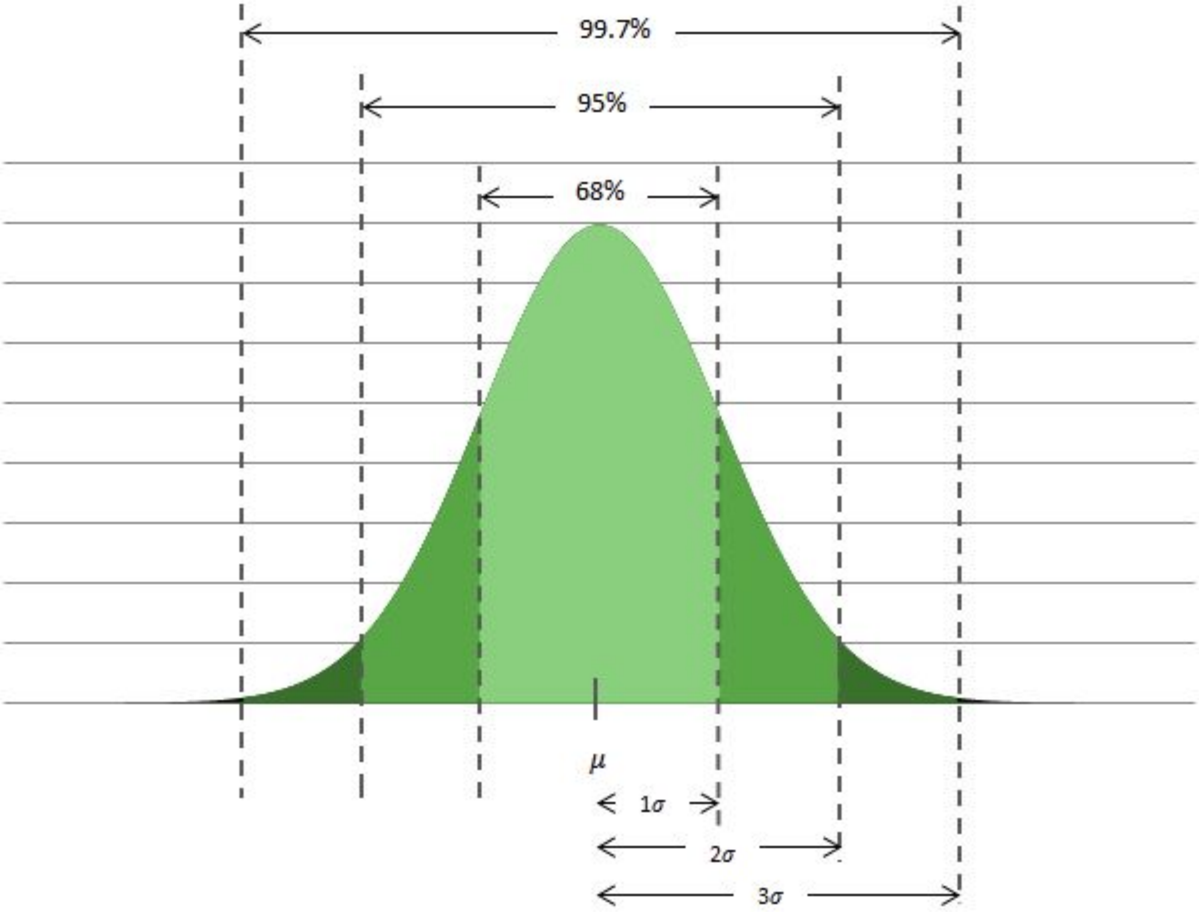
Normal Distribution - 68-95-99.7 Rule

For a normal distribution with mean , and standard deviation .

- ▷ 68% of the observations fall within one standard deviation of the mean
- ▷ 95% of the observations fall within two standard deviations of the mean
- ▷ 99.7% of the observations fall within three standard deviations of the mean



Normal Distribution - 68 , 95, 99.7 Rule



Normal Distribution Example

▷ Let's assume that the distribution of birth weights is normally distributed with a mean of 3500 grams with a standard deviation of 500 grams.

▷ What proportion of infants weigh between 3000 and 4000 grams at birth?

▷ What proportion weigh between 2500 and 4500 grams?

▷ What percentage weigh more than 4000 grams?

Normal Distribution Example

- Let's assume that the distribution of birth weights is normally distributed with a mean of 3500 grams with a standard deviation of 500 grams.

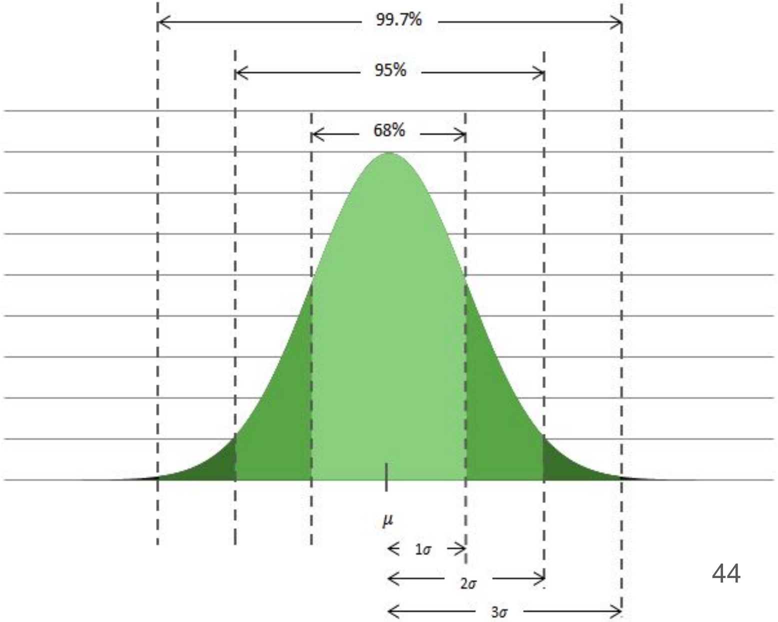
- What proportion of infants weigh between 3000 and 4000 grams at birth?

- What proportion weigh between 2500 and 4500 grams?

- What percentage weigh more than 4000 grams?

- 68% rule tells us that 68% of data is within ($\mu - \sigma$) and ($\mu + \sigma$)

- 95% rule tells us that 95% of data is within ($\mu - 2\sigma$) and ($\mu + 2\sigma$)



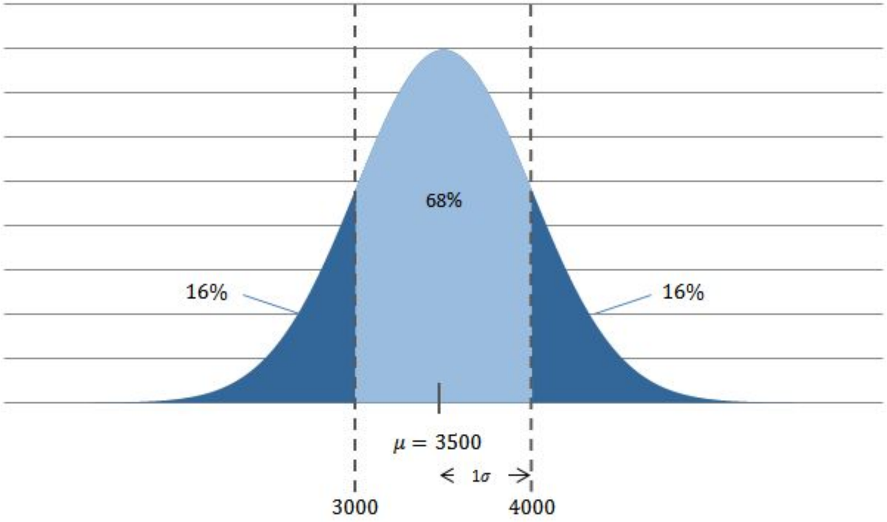
Think and Answer!

▷ What proportion is between 3000 and 4000 grams at birth?

Apply 68% rule, 3500 − 500 = 3000 grams and 3500+500=4000 grams

▷ What proportion weigh between 2500 and 4500 grams?

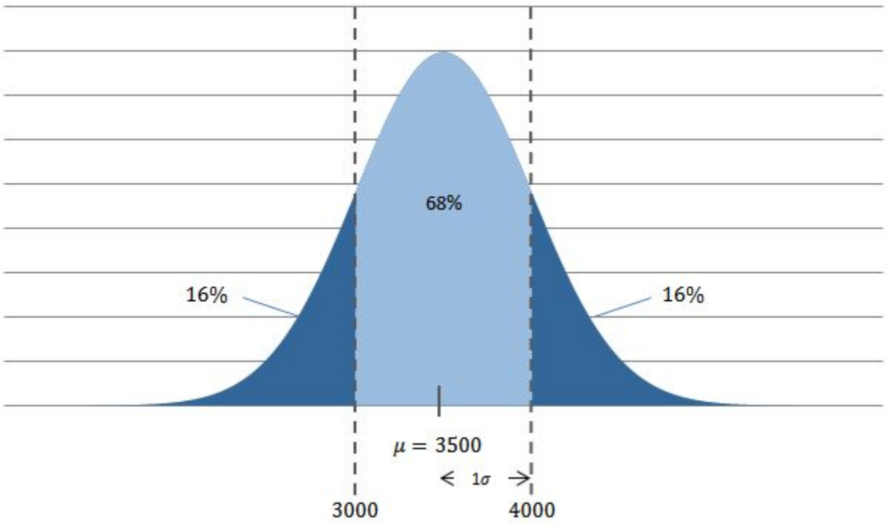
95% rule, 3500 − 2 * 500 = 2500 and 3500 + 2 * 500 = 4500



Think and Answer!

▹ What percentage weigh more than 4000 grams?

68% between 3000 grams and 4000 grams. 100%−68%=32% of the data are outside
Considering the symmetry, (16%) is below 3000 grams and (16%) is above 4000 grams.



Exercise

The distribution of SAT scores for the verbal section for high school seniors is approximately a normal distribution with a mean of 504 and a standard deviation of 111.

What proportion of seniors score between 393 and 615?

Answer

μ= 504, σ = 111

393 − 504 = −111

615 − 504 = 111

393 and 615 are one standard deviation away from the mean of 504

393 = 504 − 111 = $\mu - \sigma$

615 = 504 + 111 = $\mu + \sigma$

68% rule tells us that 68% of scores are between 393 and 615.

Normal Distribution

The normal distribution with mean μ and variance σ^2 is often referred to as $N(\mu,\sigma^2)$

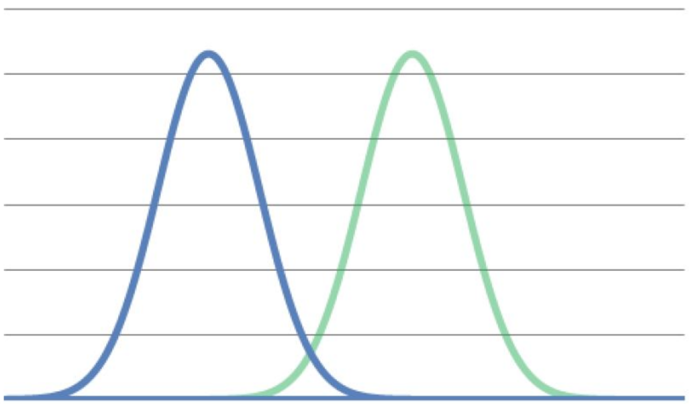
When a random variable X is distributed normally with mean μ and variance σ^2, one may write

$$X \sim N(\mu,\sigma^2)$$

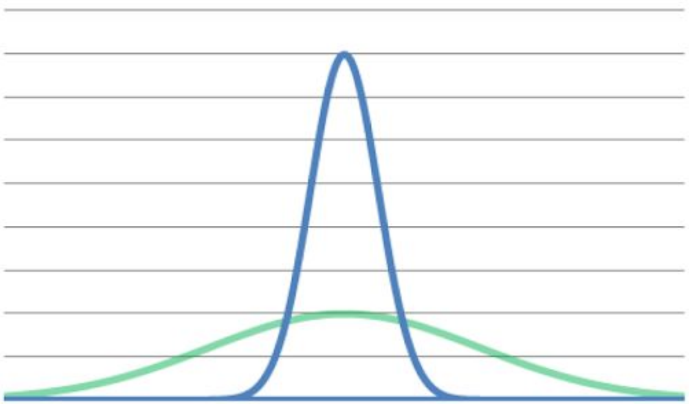
The above notation means also that X is sampled from $N(\mu,\sigma^2)$

Normal Distribution

The mean is the center of the distribution and is the point that splits the area under the bell shaped curve in half.



Different Means



Different Variance

Module1/ Lecture2

Table of contents

Standard Normal Distribution

Standardized Normal Distribution

▷ If we measure in standard deviation units, all normal distribution density curves are the same.

▷ If we define an area under a normal distribution in terms of standard deviation units, then regardless of the actual values of the mean and standard deviation, that area corresponds with the same proportion of observations.

▷ Convert a value into standard deviation units by calculating its Z-score

▷ Z-score tells us how many standard deviation x is from the mean

$$z = \frac{x - \mu}{\sigma}$$

Standardized Normal Distribution

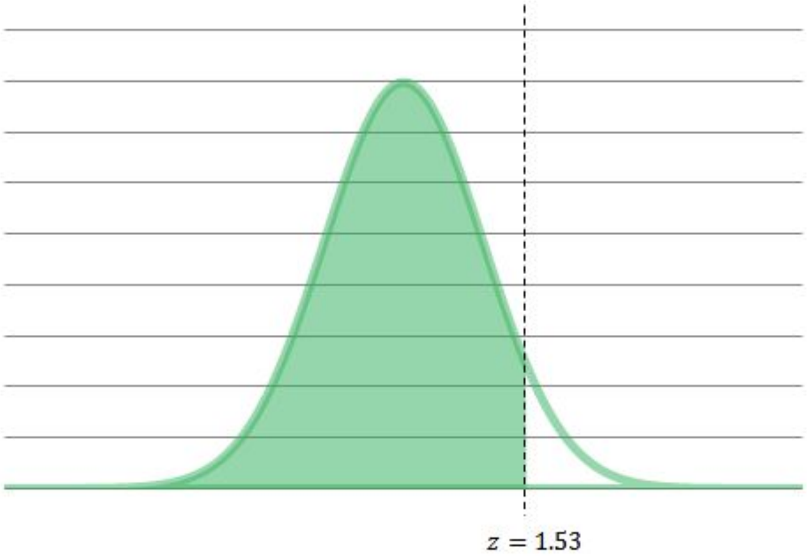
▷ If the original variable was normally distributed with a specific mean standard deviation, the distribution of standardized z values follows a normal distribution with a **mean of 0** and a **standard deviation of 1** (the standard Normal distribution).

▷ This means that regardless mean and standard deviation of a particular normal distribution, **we can ”transform” it a standard Normal distribution.**

▷ We can use a single distribution (the standard Normal distribution) to compute areas under the curve (which translate into proportions of observations).

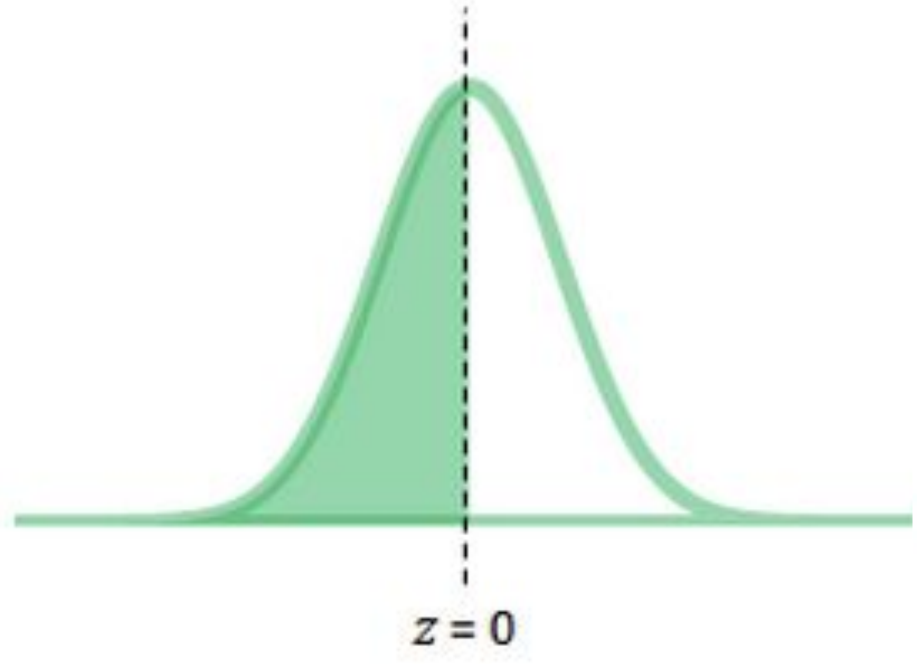
Standardized Normal Distribution

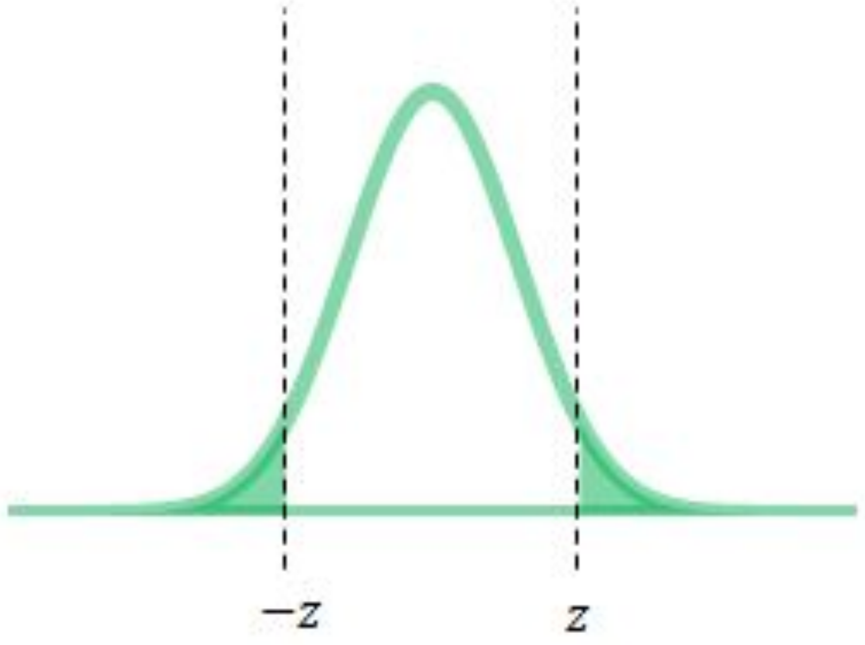
- Convert a value into standard deviation units by calculating its Z-score
- Z-score tells us how many standard deviation x is from the mean



The area under the standard Normal curve to the left of z is 0.9370.

Standardized Normal Distribution

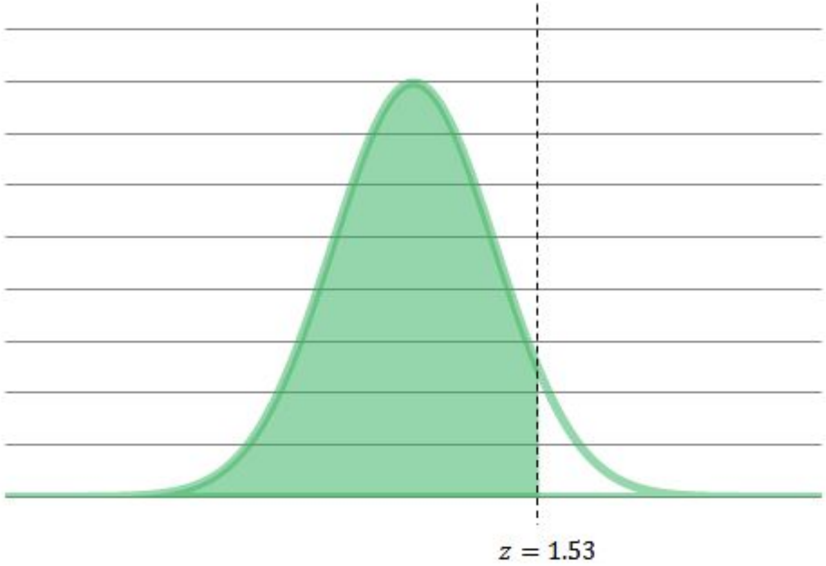




Standardized Normal Distribution

- Convert a value into standard deviation units by calculating its Z-score
- Z-score tells us how many standard deviation x is from the mean

$$z = \frac{x - \mu}{\sigma}$$



The area under the standard Normal curve to the left of z is 0.9370.

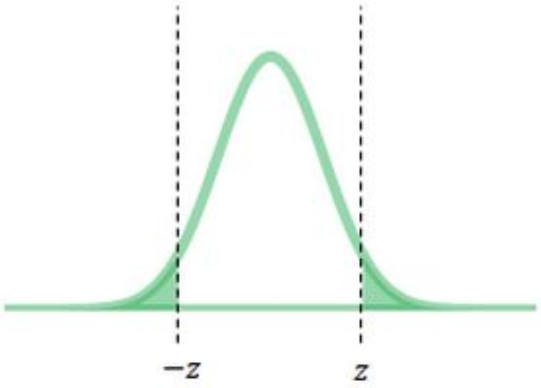
Properties of Standardized Normal Distribution

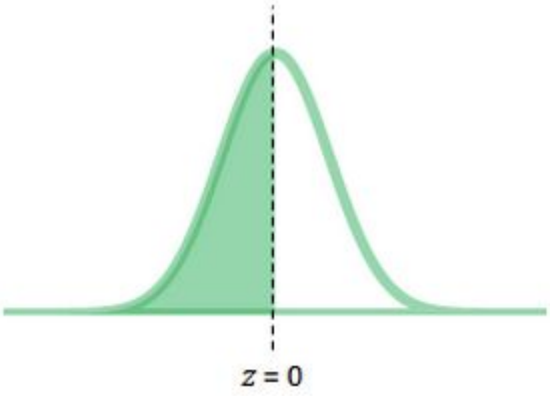
▹ The total area under the standard normal curve is 1.00.

▹ The curve is perfectly symmetrical. If z is greater than 0, then the probability that a given observation is less than -z is equal to the probability that a given observation is greater than z.

▹ The standard normal curve is centered on 0.

▹ The probability that z C a is equal to the probability that z > a. There is no area under the curve for a single point. Similarly, the probability that z ≤ a is equal to the probability that z < a.

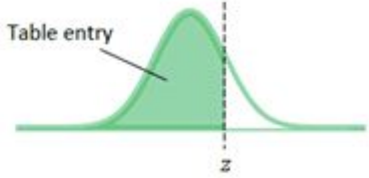




Standard Normal Distribution Probabilities Table

Table A. Standard Normal Probabilities (continued)

Table entry for z is the area under the standard Normal curve to the left of z.

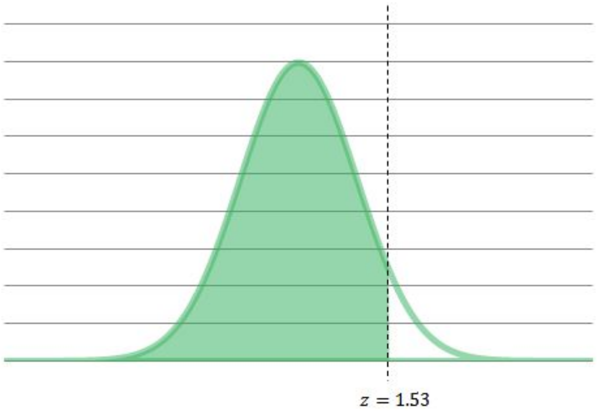


z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.00	0.5000	0.5040	0.5080	[illegible]	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.10	0.5398	0.5438	0.5478	[illegible]	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.20	0.5793	0.5832	0.5871	[illegible]	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.30	0.6179	0.6217	0.6255	[illegible]	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.40	0.6554	0.6591	0.6628	[illegible]	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.50	0.6915	0.6950	0.6985	[illegible]	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.60	0.7257	0.7291	0.7324	[illegible]	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.70	0.7580	0.7611	0.7642	[illegible]	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.80	0.7881	0.7910	0.7939	[illegible]	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.90	0.8159	0.8186	0.8212	[illegible]	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.00	0.8413	0.8438	0.8461	[illegible]	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.10	0.8643	0.8665	0.8686	[illegible]	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.20	0.8849	0.8869	0.8888	[illegible]	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.30	0.9032	0.9049	0.9066	[illegible]	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.40	0.9192	0.9207	0.9222	[illegible]	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.50	[illegible]	[illegible]	[illegible]	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.60	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.70	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.80	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.90	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767

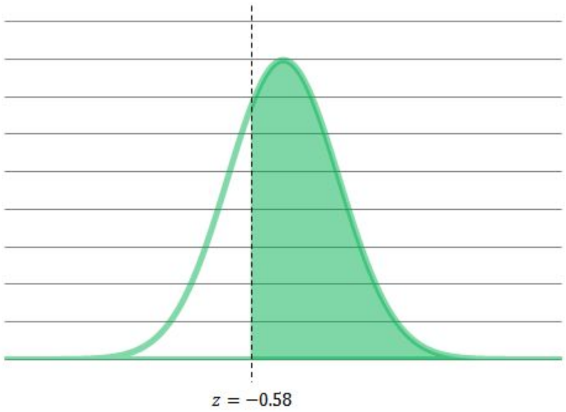
Exercise

Using the tables,

- ▷ find the area under the standard normal curve to the left of $z = 1:53$
- ▷ find the proportion of observations greater than $z = -0:58$



The area under the standard normal curve to the left of z is 0.937.



The area under the standard normal curve to the right of z is $1 - 0:281 = 0:719$.

Steps to solve the question

1. State the problem in terms of x.

2. Standardize x to restate the problem in terms of the standard normal variable z. Draw a picture to show the area under the standard normal curve.

3. Find the required area under the standard normal curve using Table and keep in mind that the total area under the curve is 1.

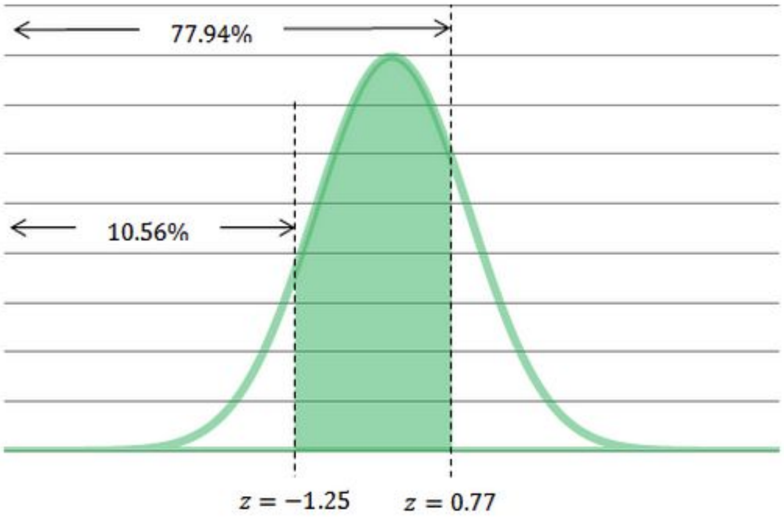
Exercise

Using the tables, find the area under the standard normal curve between z=-1.25 and z=0.77.

The area to the left of z = −1:25 is 0.1056.

The area to the left of z = 0:77 is 0.7794.

The area between z = −1:25 and z = 0:77 is equal to 0:7794−0:1056 = 0:6738.



R Functions - Normal Distribution

pnorm() computes the probability that a normally distributed random number will be less than the given number. This function is also called the ”Cumulative Distribution Function” (CDF).

Calculate the area to the left of z：$P(Z \leq z)$

```
> pnorm(z)
```

For non-standardized normal distribution

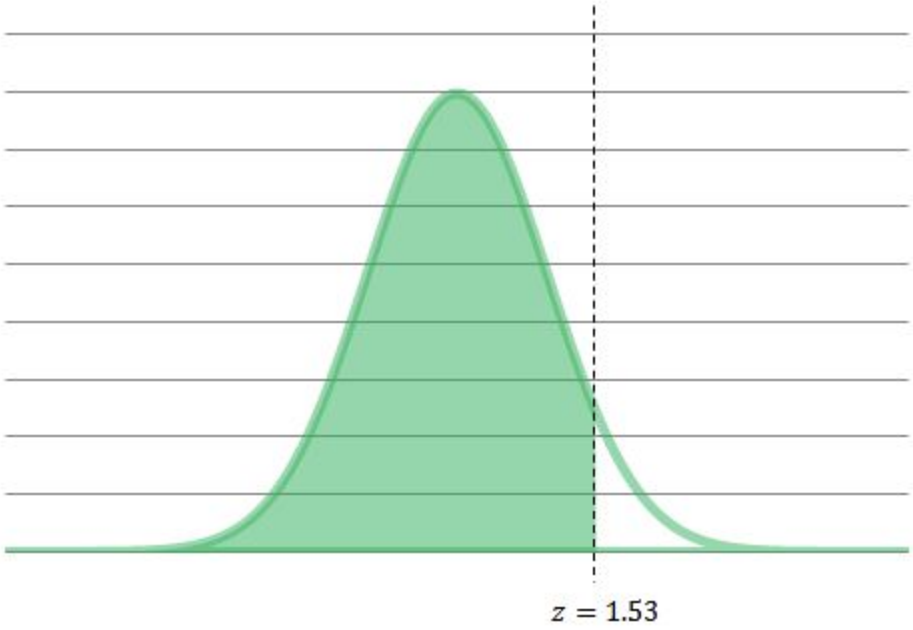
```
pnorm(x, mean=a, sd=b) # calculate the area to the left of x
```

Find the area under the standard curve

z=1.53

R command:

```
> pnorm(1.53)
0.9369916
```



The area under the standard Normal curve to the left of z is 0.937.

Standardized Normal Distribution - An example

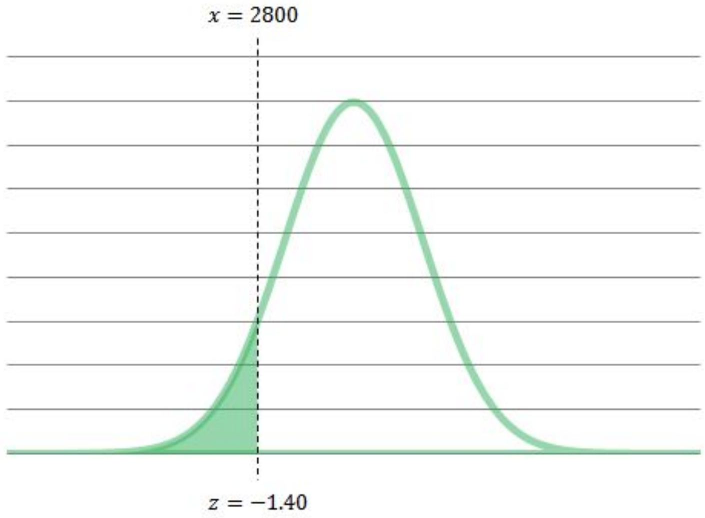
Let’s assume that the birth weights of newborns are normally distributed

with a mean of 3500g with a standard deviation of 500g.

What proportion of infants weigh less than 2800g?

The variable x, the birth weight, has the N(3500,500) distribution

```
> pnorm(2800, mean=3500, sd=500)
0.08075666
```

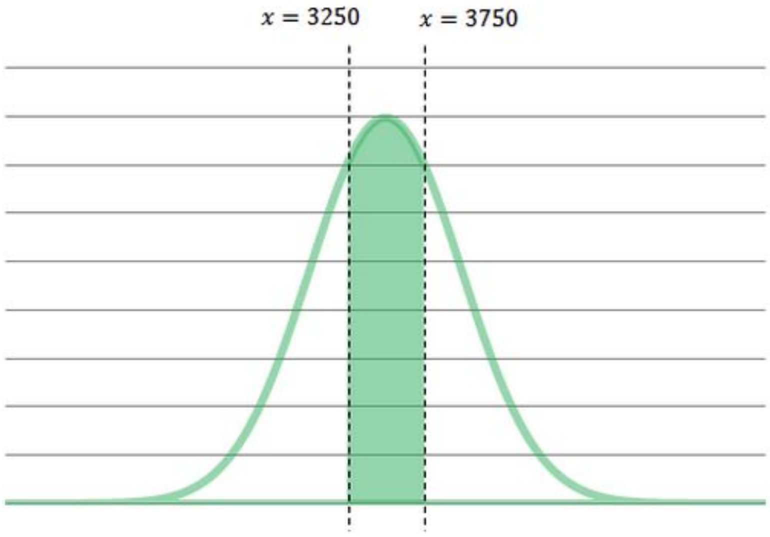


Standardized Normal Distribution – an example

Let's assume that the birth weights of newborns are normally distributed

with a mean of 3500g with a standard deviation of 500g.

What proportion of infants weigh between 3250g and 3750g?



Example

Standardized Normal Distribution – an example

- ▹ State the problem in terms of x. Let x be the birth weight of infants.

The variable x has the N(3500,500) distribution. We are interested in the proportion of infants with $3250 < x < 3750$ grams.

- ▹ Standardize x to restate the problem in terms of the standard normal variable z.

- ▹ Find the required area under the standard normal curve using the Table. The area between $z = -0{:}50$ and $z = 0{:}50$ is

$0{:}6915 - 0{:}3085 = 0{:}3830$

R command

```
> pnorm(3750, mean=3500, sd=500)-pnorm(3250, mean=3500, sd=500)
0.3829249
```

R Functions on Normal Distribution

dnorm() - Given a set of values it returns the height of the probability distribution at each point. If you only give the points it assumes you want to use a mean of 0 and standard deviation of 1.

Normal Distribution Probability density function (PDF)

$$p(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

For standard normal distribution we have $\mu = 0$, $\sigma = 1$

$$p(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

```
density_standard_norm <- function(x){1/sqrt(2*pi)*exp(-0.5*x^2)}
> dnorm(0)
0.3989423
> dnorm(0, mean=4, sd=10)
0.03682701
> pnorm(0)
0.5
```

R Function on Normal Distribution - qnorm

In statistics, quantiles are cut points dividing the range of a probability distribution into contiguous intervals with equal probabilities.

Given a probability p and a distribution, we want to calculate the corresponding quantile for p: the value x such that P(X x)=p

```
> qnorm(x)
For non-standardized normal distribution
> qnorm(x, mean=a, sd=b)
```

R Function on Normal Distribution

Density, distribution function, quantile function and random generation for the normal distribution with mean equal to mean and standard deviation equal to sd.

Usage:

```
dnorm(x, mean = 0, sd = 1, log = FALSE)
pnorm(q, mean = 0, sd = 1, lower.tail = TRUE, log.p = FALSE)
qnorm(p, mean = 0, sd = 1, lower.tail = TRUE, log.p = FALSE)
rnorm(n, mean = 0, sd = 1)
```

To read the manual

```
> help(pnorm)
```

Outliers

Outliers

Outliers are data points that do not fit with the general pattern of the data. They can be detected using graphical summaries.

Quartiles divide a rank-ordered data set into four equal parts. The values that divide each part are called the first, second, and third quartiles; They are denoted by Q1, Q2, and Q3, respectively.

▷ Q1 is the ”middle” value in the first half of the rank-ordered data set.

▷ Q2 is the median value in the set.

▷ Q3 is the ”middle” value in the second half of the rank-ordered data

set.

IQR (Inter-Quartile Range) = Q3 - Q1

It is a measure of variability. Outliers are defined as any points that are

$\leq Q1 - 1{,}5 \times IQR$ OR $\geq Q3 + 1{,}5 \times IQR$

Outliers - An Example

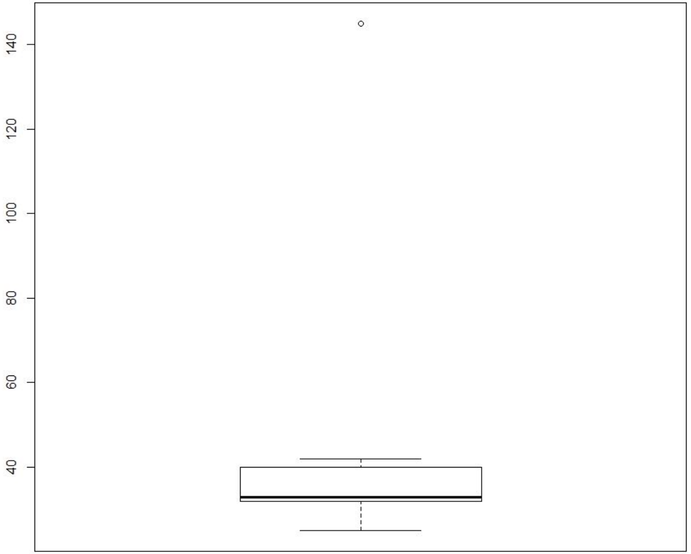
The earnings (in thousands of dollars) for 9 people are:

$\langle 35; 40; 145; 33; 30; 42; 32; 32; 2; 5 \rangle$

```
> earning <- c(35, 40, 145, 33, 30, 42, 32, 32, 25)
> summary(earning)
Min.    1st Qu.   Median   Mean    3rd Qu.   Max.
25      32        33       46      40        145
> boxplot(earning)
```

Boxplot - Showing Outliers

boxplot() function in R can detect outliers and visualize it.



Boxplot

Sampling

Sampling

▷ The population: entire group of individuals or things that we want information about or to answer questions about.

▷ A Sample: a set of data collected and/or selected from a statistical population by a defined procedure.

The elements of a sample are known as *sample points, sampling units or observations*.

Often we gather information about only part of the population (sample) and use this information to make educated guesses about qualities of the larger group (population).

Selection Bias

Biases introduced by sampling are often termed selection biases.

▷ To avoid them, samples by chance .

▷ Ensuring that each individual or element from the population has an equal chance of being selected is key to protecting against selection bias.

▷ Our data collection method is critically important to our ability to analyze data correctly and to make inferences and conclusions.

▷ We should be mindful of sampling methods and potential for bias as we critically analyze our results.

Inference about a population

A parameter is a number that describes a population.

A statistic is a number that is computed from the sample data.

Typical notation for population parameters and their corresponding sample statistic:

	Population Parameter	Sample Statistic
Mean	μ ("mu")	$\bar{x}$ ("x bar")
Variance	σ^2 ("sigma squared")	s^2
Standard Deviation	σ ("sigma")	s
Proportion	p	$\hat{p}$ ("p hat")

Central Limit Theorem

The Central Limit Theorem – An Example

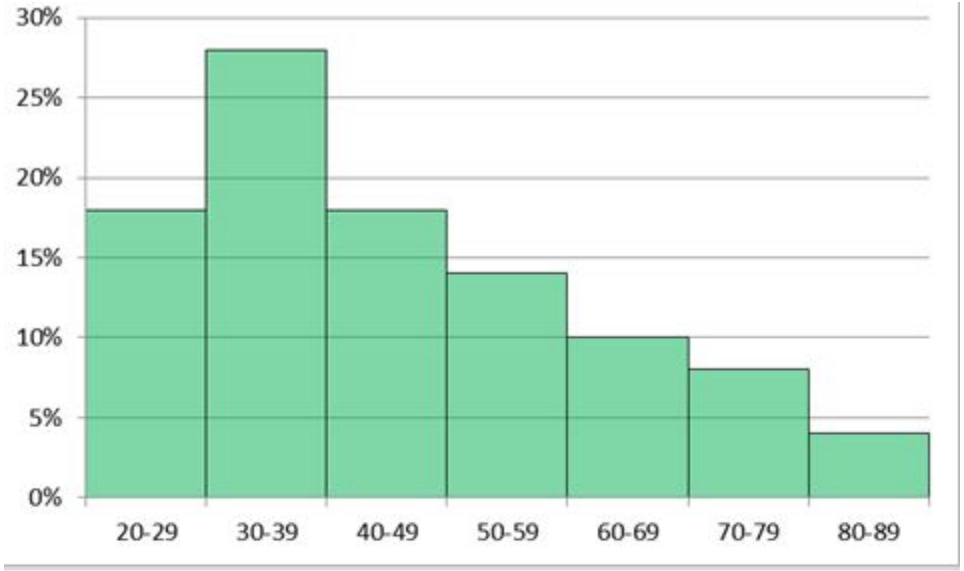
We would like to estimate the mean age of employees at company A. Let's assume that the population is made up of only 50 individuals with the following age distribution:

42	20	32	47	31
66	25	64	25	46
76	56	32	20	50
60	58	31	83	51
22	32	64	49	75
40	43	54	44	62
46	27	32	49	37
38	59	33	59	73
26	26	83	71	39
35	33	35	28	35

The population mean of the ages of employees is 45.28 years

The Central Limit Theorem – An Example

We would like to estimate the mean age of employees at company A. Let's assume that the population is made up of only 50 individuals with the following age distribution:



The population mean of the ages of employees is 45.28 years

The Central Limit Theorem – An Example

Assume we take two random samples of size 5. The resulting ages in each of the two samples.

Sample 1: 20,44,46,20,44

Sample 2: 83,32,31,50,32

- ▹ The first sample has a sample mean of 34.6 and a sample median of 44.
- ▹ The second sample has a sample mean of 45.6 and a sample median of 32.

Neither the sample mean nor the sample median will always fall closer to the population mean in a given sample.

To evaluate each of these sample statistics and their ability to estimate the true population value, we must not rely on just one example (one sample).

We'd like to compare the distribution of the sample mean and sample median if we take hundreds of random samples of the same size.

The Central Limit Theorem – An Example

- We want to evaluate how well the sample mean and sample median perform as estimators of the population mean.

- Using a computer to randomly select 10,000 samples of size 5.

Sample	x_1	x_2	x_3	x_4	x_5	$\bar{x}$	m
1	20	44	46	20	44	34.6	43
2	83	32	31	50	32	45.6	32
3	58	49	31	32	50	44	49
4	49	38	31	71	32	44.2	38
5	40	27	76	47	62	50.4	47
6	59	27	32	66	46	46	46
7	31	64	38	42	62	47.4	42
8	83	49	50	39	22	48.6	49
9	40	26	22	49	54	38.2	40
10	27	47	64	39	54	46.2	47
...							
10,000	25	26	33	60	71	43	33

The Central Limit Theorem – An Example

When the number of samples taken from a population is sufficiently large, the sampling distribution of the sample mean, will be approximately normally distributed with an expected value of μ and a standard deviation of σ.

Say you take a random sample of size n from a population with mean μ and standard deviation σ.

The larger the sample size,

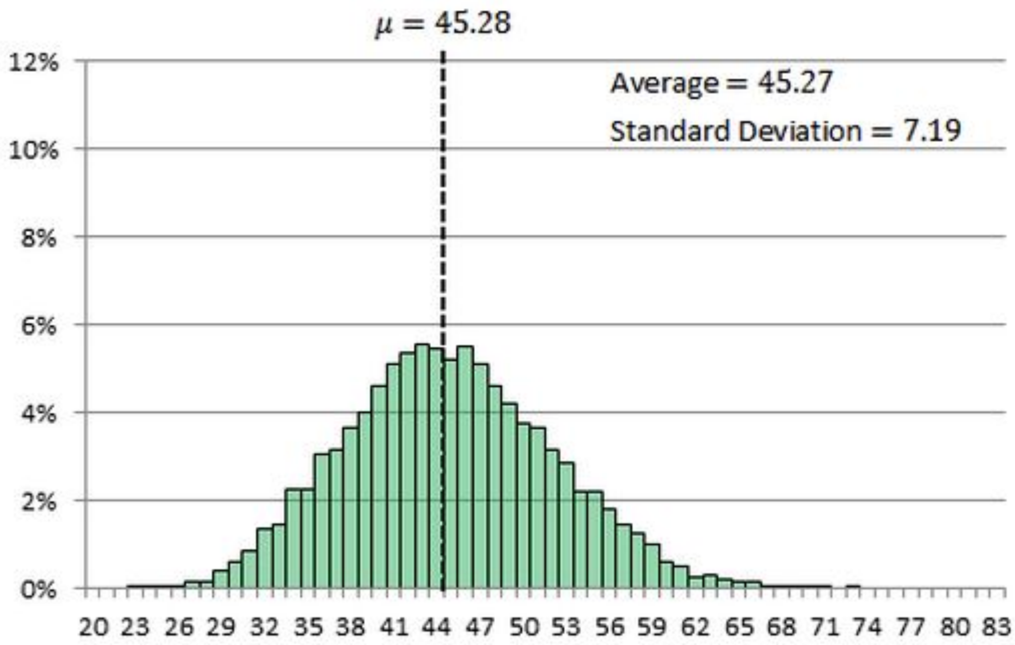
1. the closer the sampling distribution of the sample means will be to the normal distribution and

2. the smaller the variance of the sample mean.

The Central Limit Theorem – An Example

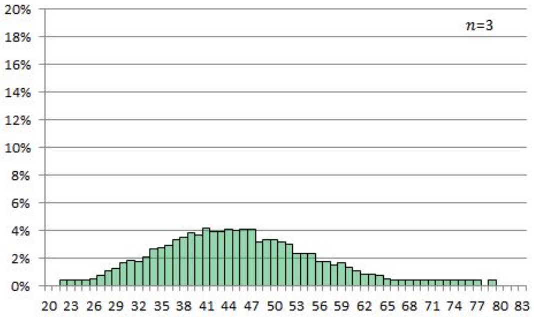
The distribution of the sample mean x̄ for 10000 samples of size 5.

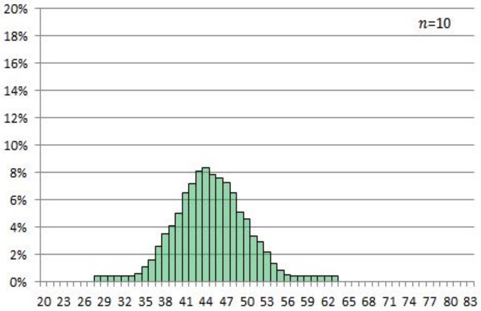
The distribution of the Sample mean is centered over the true value and has less spread or variability than the distribution of the sample mean.

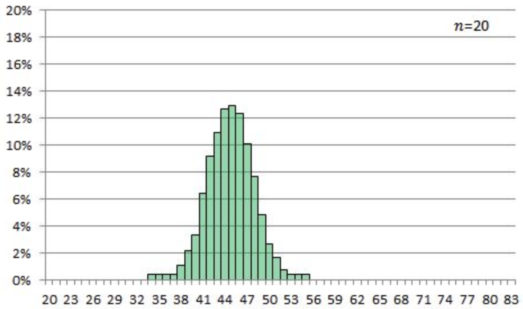


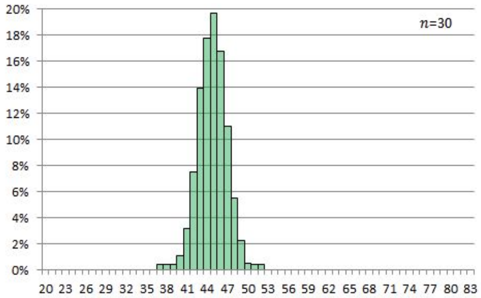
The Central Limit Theorem – An Example

The population has a mean of 45.28 and a standard deviation of 17.20









n	$\bar{x}_{\bar{x}}$	$\sigma_{\bar{x}}$
3	45.25	9.57
5	45.27	7.19
10	45.20	4.80
15	45.28	3.72
20	45.25	2.97
25	45.28	2.43
30	45.28	1.97

Try an Online Simulation

For better understanding try the following online demo

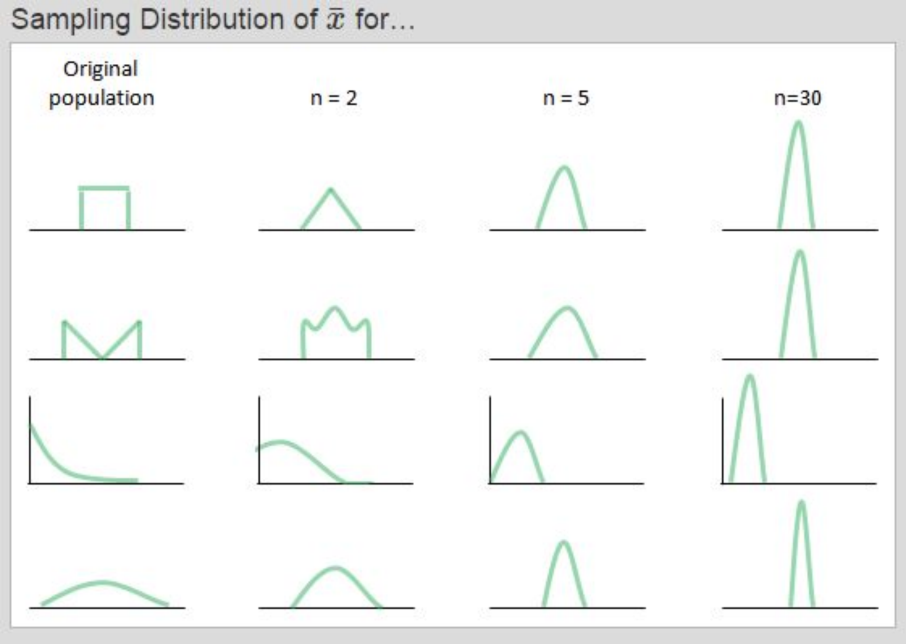
https://gallery.shinyapps.io/CLT_mean/

Change the parameters and scroll down to see the “Sampling Distribution”

How large should n be?

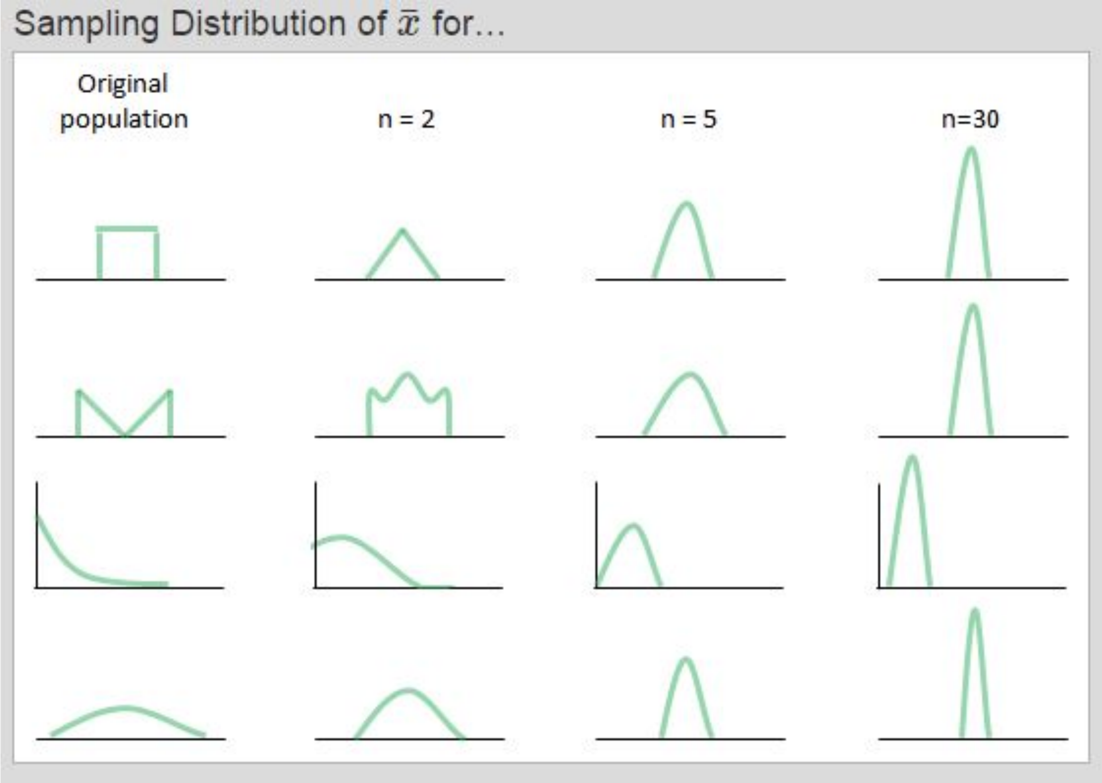
If the underlying population is approximately normal, then even small values of n will give a sampling distribution of sample means that are normally distributed.

Generally, the rule of thumb is that n should be $n \geq 30$ for the distribution of the sample means to be reasonably normally distributed.



Skewed population distributions

For more skewed population distributions, n must be larger before the sampling distribution is sufficiently normally distributed.

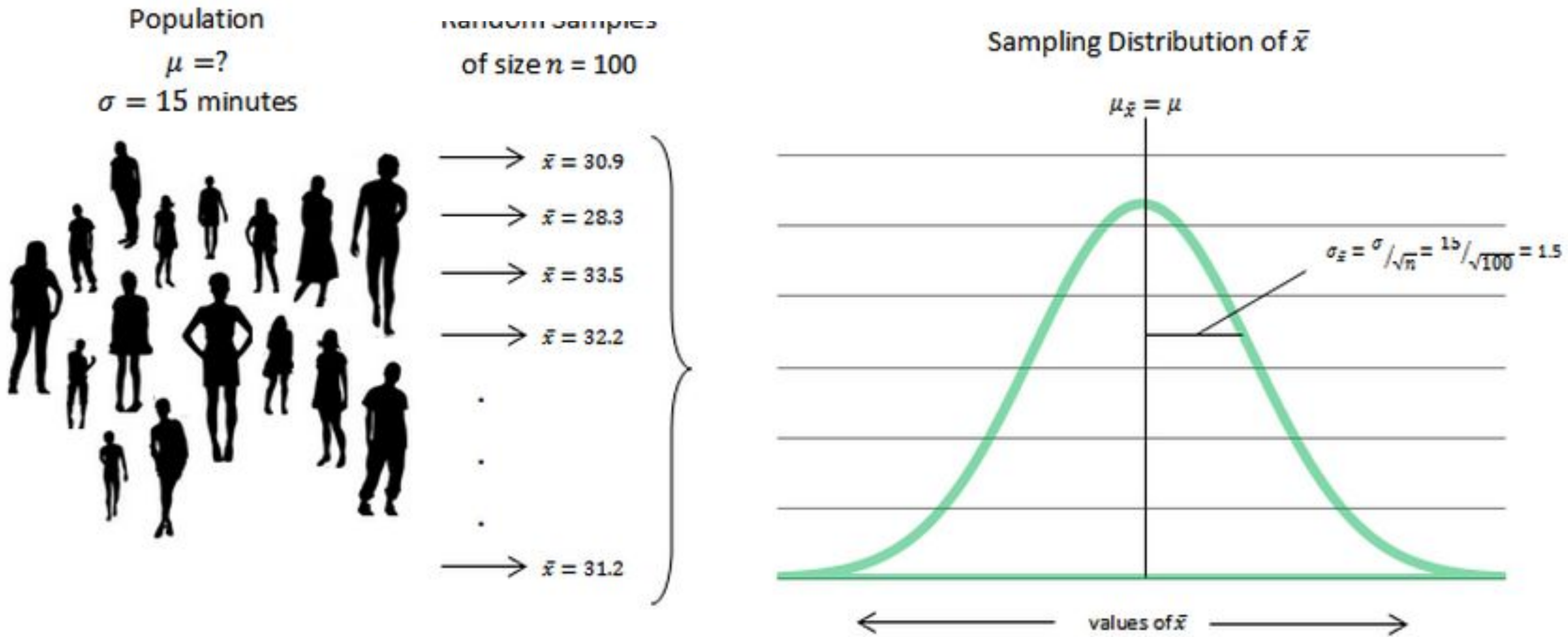


Standard Deviation of Sample Mean

If n ≥ 30, we know that the sample mean is approximately normally

distributed with Standard Error(SE).

$$\mu = \mu_{\bar{x}} \qquad SE = \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \qquad \textbf{Standard Error}$$



Exercise

Suppose we have selected a random sample of n = 36 from a

population with a mean of 80 and a sd of 6.

Find the probability that the sample mean will be between 79 and 81.

Answer

The Central Limit Theorem tells us that regardless of the shape of the underlying population, the sampling distribution of x̄ is approximately normal when n ≥ 30.

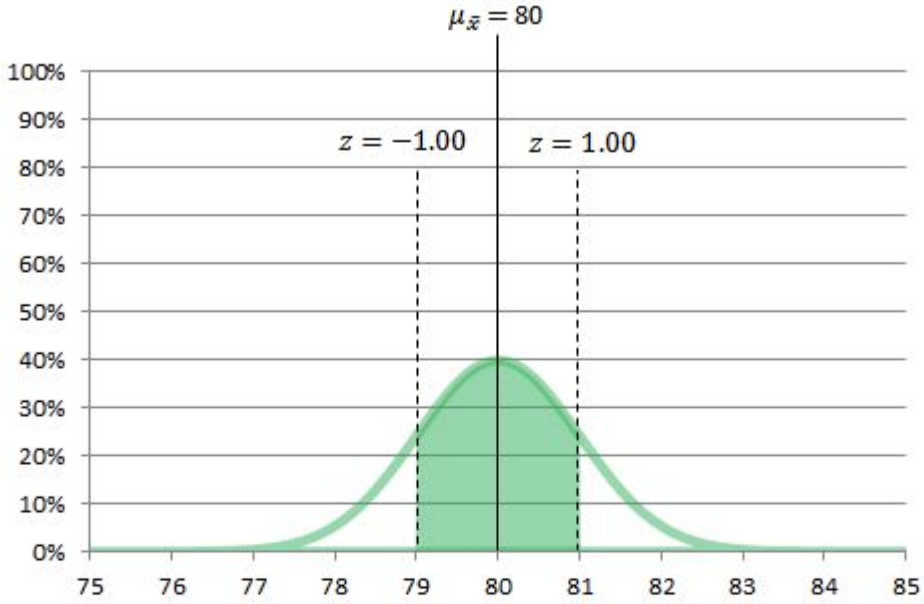
The sampling distribution will have a mean of $\mu_{\bar{x}}$ = 80 and $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{6}{6} = 1$

We first standardize the distribution and then calculate the probabilities:

$$79 < x < 81$$

$$\frac{79-80}{1} < \frac{x-80}{1} < \frac{81-80}{1}$$

$$-1 < z < 1$$

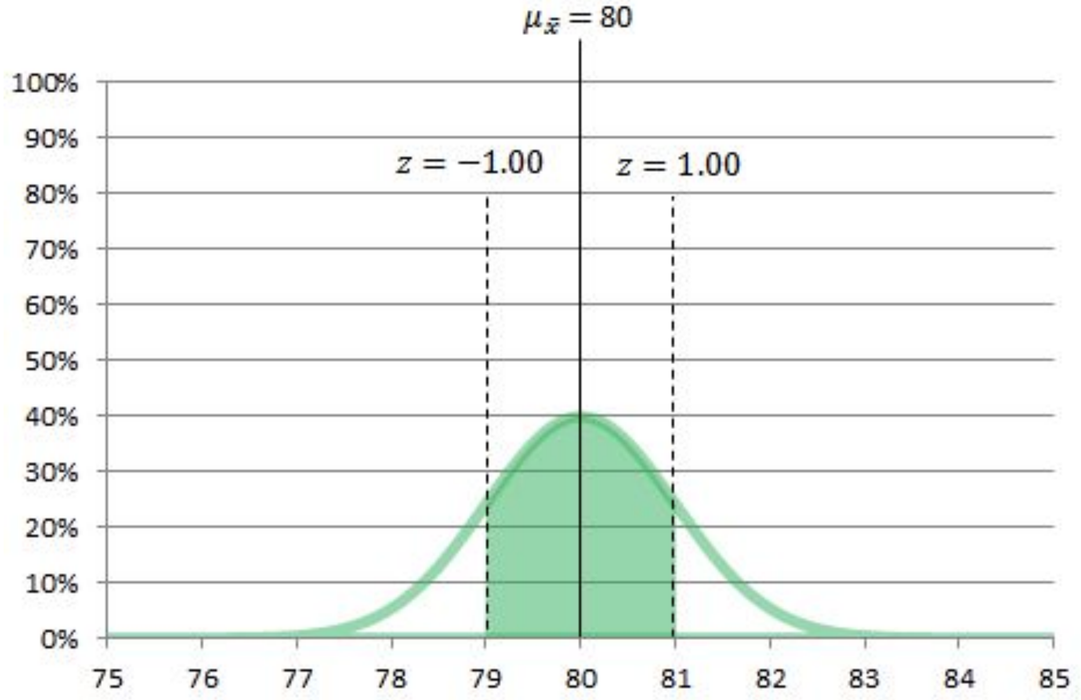


0.8413 - 0.1587=0.6826 OR
0.8413 - 0.1587=0.6826
68.26%

Answer - Using R

```
> 100 * (1 - 2 * pnorm(81, mean= 80, sd= 6/6 , lower.tail = F))

[1] 68.26895
```



Probability Density Function

A PDF is a function that computes the relative probability of an event.

For Normal Distribution, we have

$$f_{Normal}(x|\mu, \sigma) = \sigma^{-1}(2\pi)^{-\frac{1}{2}} e^{-\frac{1}{2}(x-e)^2\sigma^{-2}}$$

- ▷ A PDF can be used to calculate the probability of a range of points (or events)
- ▷ $\int_a^b f(x)dx$ is the probability we see a value in range a to b

More Background knowledge:
https://en.wikipedia.org/wiki/Normal_distribution
https://en.wikipedia.org/wiki/Central_limit_theorem

Answer

The Central Limit Theorem tells us that regardless of the shape of the underlying population, the sampling distribution of x̄ is approximately normal when n ≥ 30.

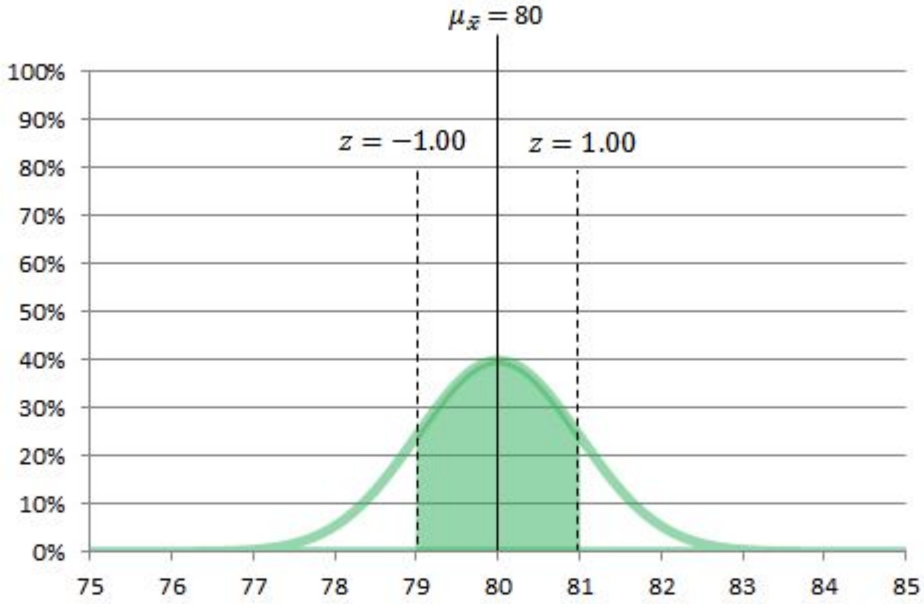
The sampling distribution will have a mean of $\mu_{\bar{x}}$ = 80 and $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{6}{6} = 1$

We first standardize the distribution and then calculate the probabilities:

$$79 < x < 81$$

$$\frac{79-80}{1} < \frac{x-80}{1} < \frac{81-80}{1}$$

$$-1 < z < 1$$



0.8413 - 0.1587=0.6826 OR
0.8413 - 0.1587=0.6826
68.26%

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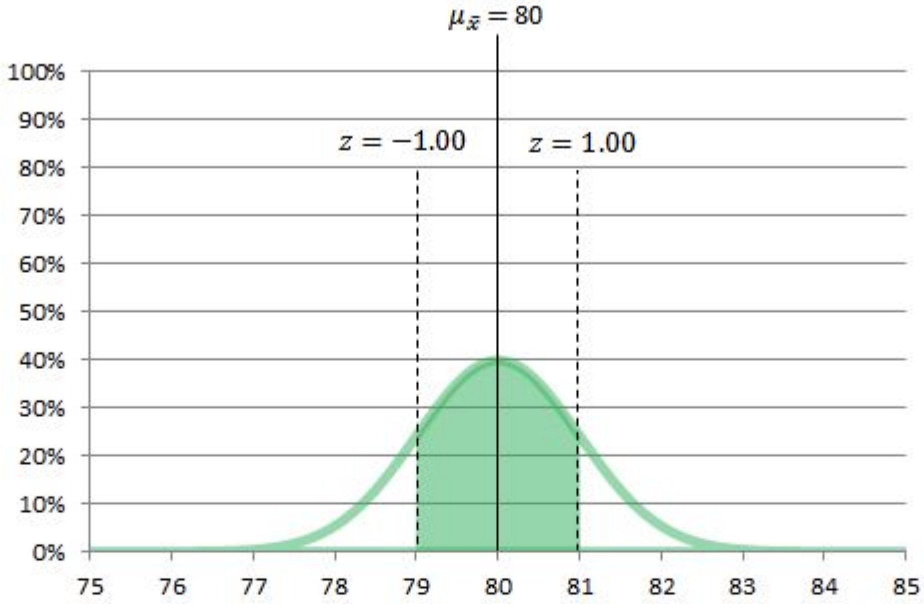

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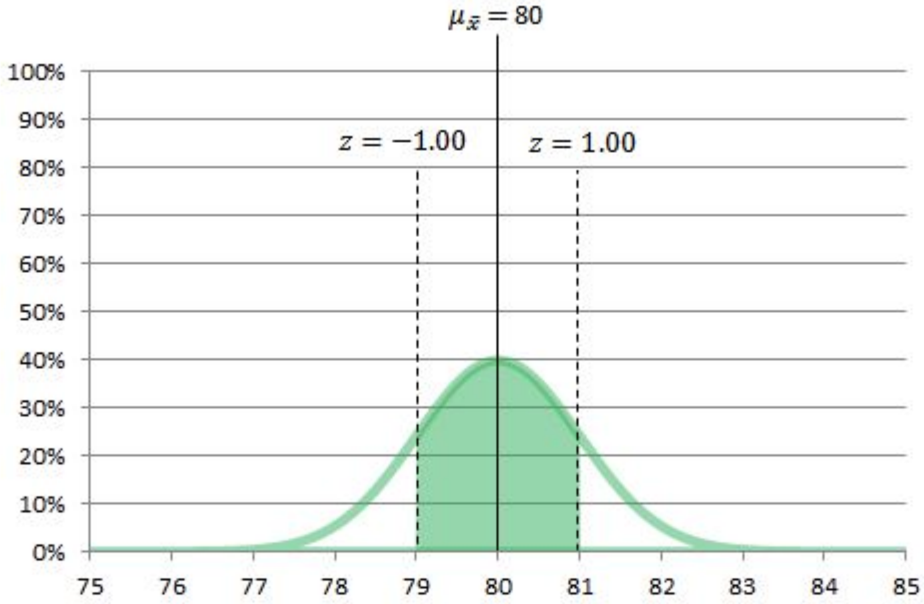

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Confidence Interval

Statistical Inference

Statistical inference provides methods for drawing conclusions about a population from sample data.

The goals of statistical inference are:

▷ Draw conclusion about a population based on sample data

▷ Provide a statement, expressed in the language of probability, of how much confidence to be placed in the conclusion

Two of the most common types of statistical inference include:

▷ Confidence Intervals for estimating the value of a population parameter

▷ Tests of Significance or Hypothesis Tests which assess the evidence for a scientific claim.

Confidence Intervals

Example - Call Center Calculating Confidence Interval

Suppose a company would like to estimate the average amount of time callers are on hold before they reach a customer service representative.

Executives from the company plan to randomly sample 100 calls and use the sample mean $\bar{x}$, to estimate population mean μ (the average of all callers wait times).

Assume that σ =15 minutes and

sample mean of the wait times is $\bar{x}$ = 30.9 minutes.

The parameter of interest is the population mean, μ.

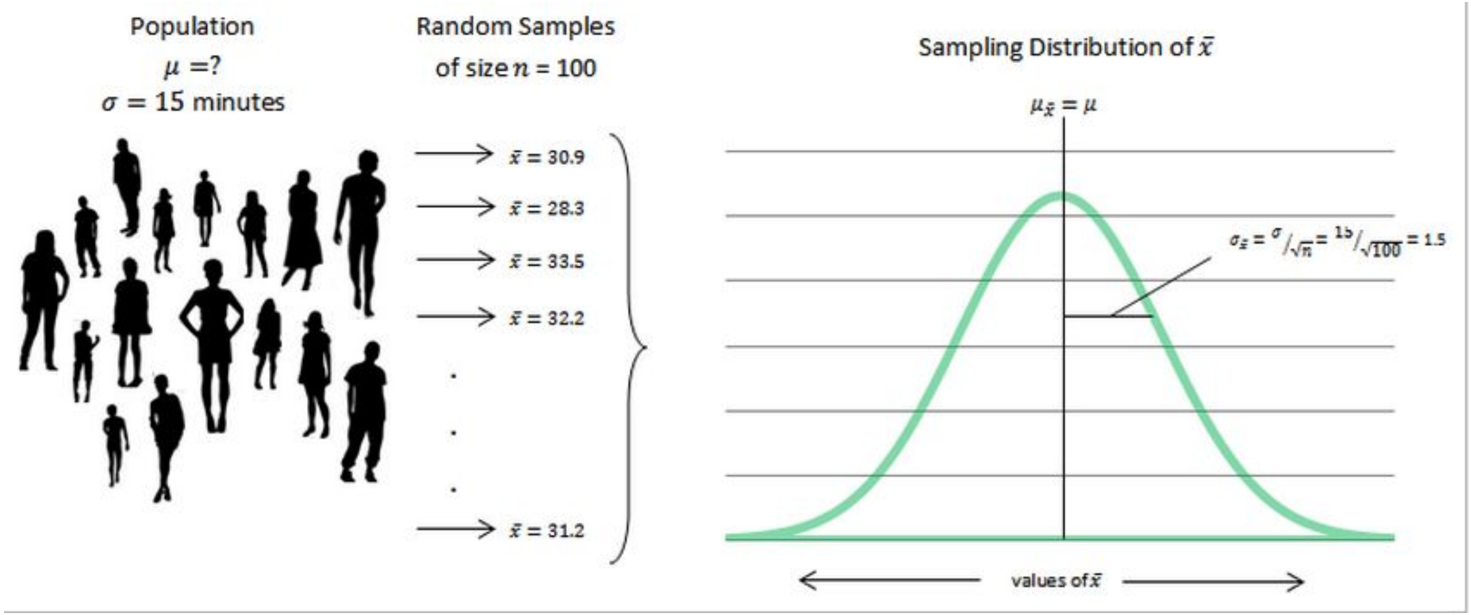
Executives also want to compute a confidence interval to assess the accuracy of the point estimate (the sample mean).

Example calculating confidence interval

Since n ≥ 30 (here, n=100), we know that the sample mean is approximately

normally distributed with a mean $\mu_{\bar{x}}=\mu$ and a standard deviation of

$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{\sigma}{\sqrt{100}}$ = 1.5 minutes.



An example calculating confidence interval

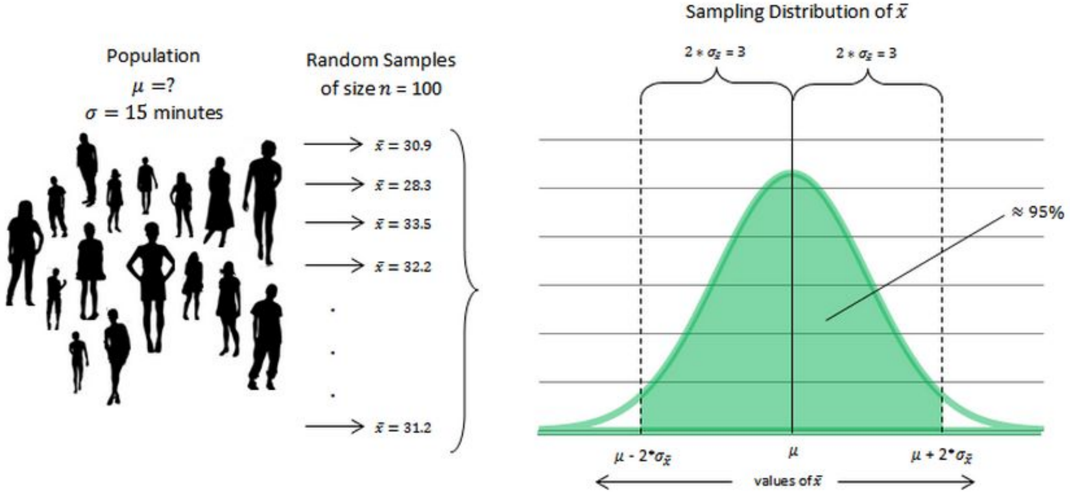
In a normal distribution, 95% of the sample means are within 2 SDs of the population mean (z = 1.96 corresponds to an area of 95% under the curve).

The sample mean, $\bar{x}$, and the population mean, μ, are within three minutes (2 SDs) of each other 95% of the time.

If our estimate of the population mean is between $\bar{x} \pm 3$

$\bar{x} - 3 = 30{:}9 - 3 = 27{:}9$ and

$\bar{x} + 3 = 30{:}9 + 3 = 33{:}9$ then we will be right 95% of the time.

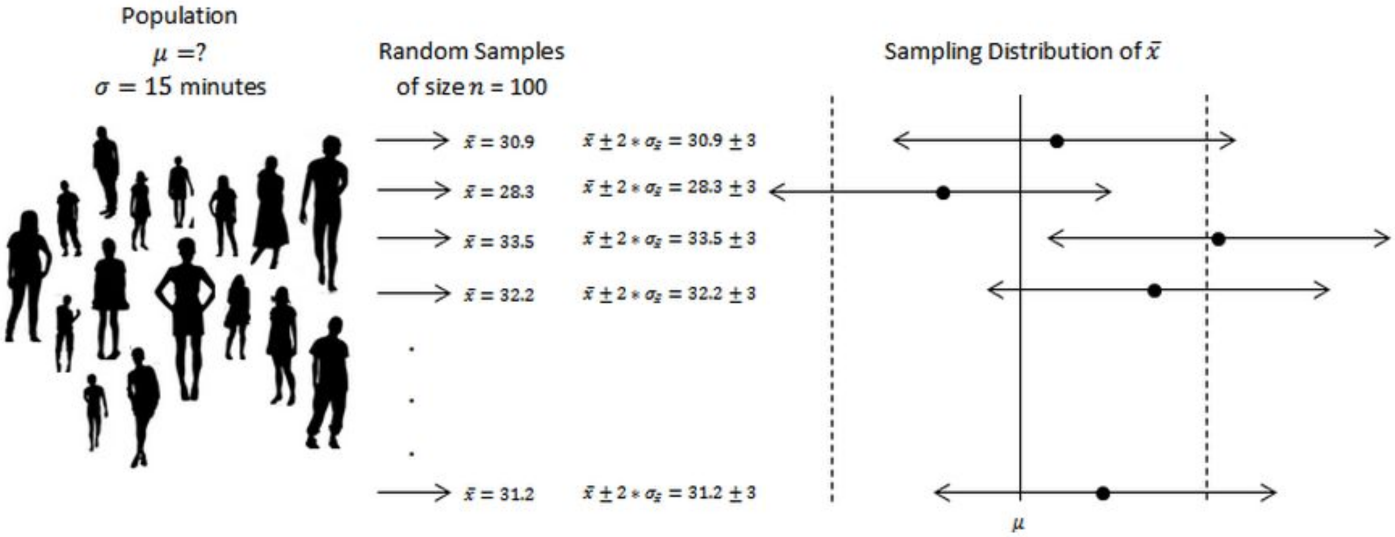


An example calculating confidence interval

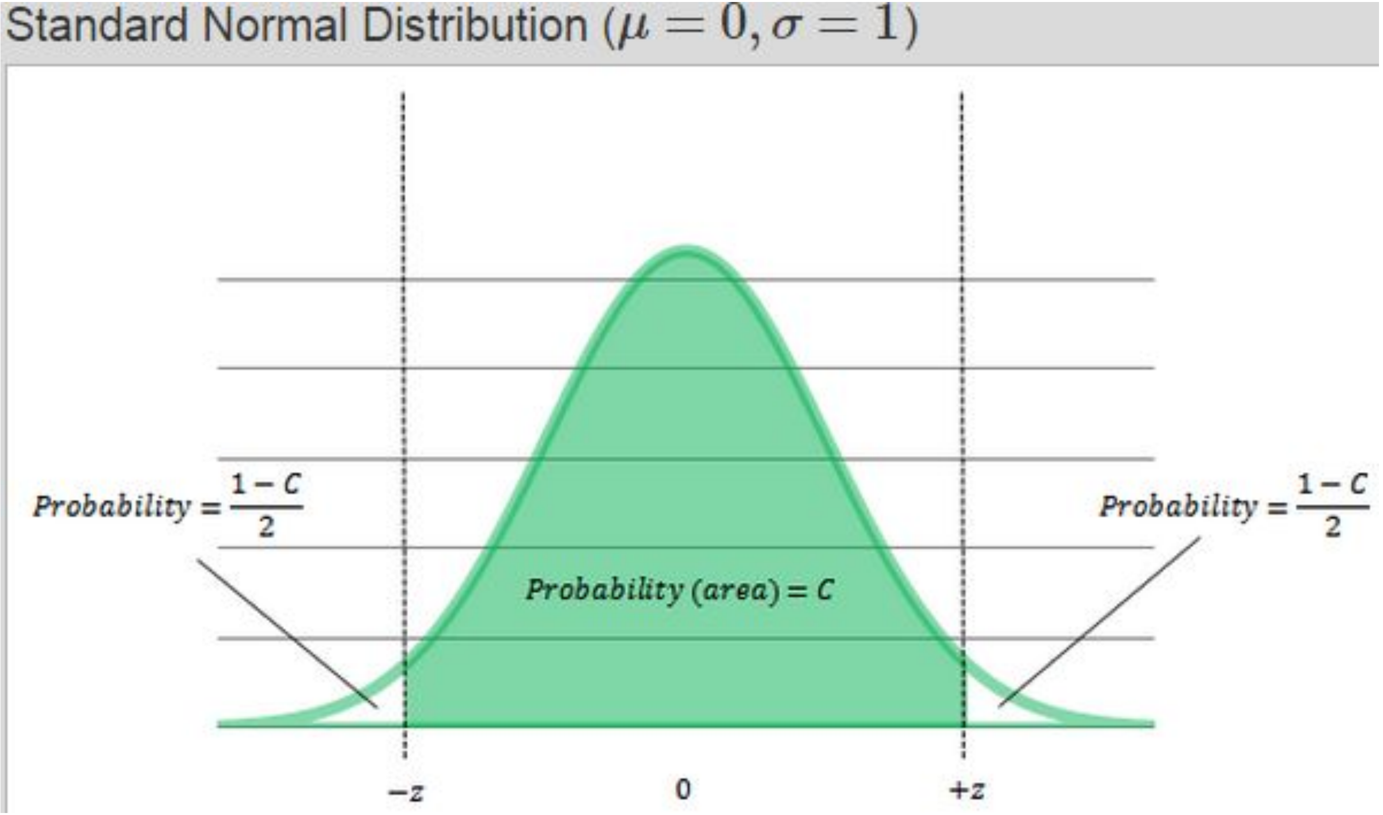
▹ We know that 95% of the sample means will be within 2 SDs of the population mean.

▹ The sample mean 2 times the SD ($\bar{x} \pm 2.\ \sigma_{\bar{x}}$) is called the 95% confidence interval for the population mean.

▹ The two-way arrows in the figure represent the confidence intervals from each random sample. The center point is the sample mean and the end of the arrows show the edges of each interval.



Confidence interval



Confidence interval

▷ Confidence intervals are of the form: **estimate ± margin of error**

▷ To calculate a confidence interval with a confidence level of C for the population mean, μ, we use the following formula:

$$\bar{x} \pm z.\frac{\sigma}{\sqrt{n}}$$

z is the appropriate critical value corresponding to the confidence level.

▷ Typical values of C with their associated values of z are:

▷ Generally, 95% con dence intervals are standard. Other common intervals are 90% and 99%.

Confidence Level, C	90%	95%	99%
Critical Value, z	1.645	1.960	2.576

Exercise 1 - Call Center Data Waiting Time

Suppose the executives would like to estimate the average amount of time callers are on hold before they reach a customer service representative by computing a 99% confidence interval. The population standard deviation of wait times, σ, is 15 minutes.

The sample mean of the wait times from the 100 random calls in the sample is 30.9 minutes.

To calculate a con dence interval with a con dence level of C for the population mean, μ, we use the formula: $\bar{x} \pm z \times \frac{\sigma}{\sqrt{n}}$

The 99% confidence interval is calculated as follows: $\bar{x} \pm z \times \frac{\sigma}{\sqrt{n}} =$

$$30.9 \pm 2.576 \times \frac{15}{\sqrt{100}} =$$

$$30.9 \pm 2.576 \times 1.5 = \{27.036, 34.764\}$$

The Sample size and the margin of error

In the case of the confidence interval for the population mean, the margin of error is

$$m = z \times \frac{\sigma}{\sqrt{n}}$$

We like our confidence intervals to be narrow (small in width) with a high amount of confidence. The precision (width) of the confidence interval decreases as:

▷ z gets smaller. Smaller values of z are associated decreased confidence levels. There is a trade off between the confidence level and the margin of error.

▷ gets smaller. When the variability of the individual observations is reduced, it becomes easier to estimate the population mean with higher precision.

▷ n gets larger. Increasing the sample size n reduces the margin of error for a fixed confidence level C.

The Sample size and the margin of error

In the case of the confidence interval for the population mean, the margin of error is equal to:

$$m = z \times \frac{\sigma}{\sqrt{n}}$$

To obtain a specific margin of error, m, for a desired confidence level, C, the number of observations needed is:

$$n = \left(\frac{z \times \sigma}{m}\right)^2$$

Exercise 2 - How many callers must they sample?

Suppose the executives would like to estimate the average amount of time callers are on hold before they reach a customer service representative by computing a 95% confidence interval.

As before, the population standard deviation of wait times, σ, is 15 minutes.

The executives would like the half width of the con dence interval to be 1.0 minutes (that is, they'd like the margin of error to be 1.0).

How many callers must they sample to get their 95% con dence interval to have the desired width?

Exercise 2 - How many callers must they sample?

How many callers must they sample to get their 95% confidence interval to have the desired width?

To obtain a specific margin of error, m, for a desired confidence level, C, we use the formula to determine the number of observations needed:

$$n = \left(\frac{z \times \sigma}{m}\right)^2 = \left(\frac{1.96 \times 15}{1}\right)^2 = 29.4^2 = 864.36$$

Note: In calculations involving sample size we always round up (instead of down) to achieve the requested margin of error. In our example we need to have samples of 865 callers

Confidence Level, C	90%	95%	99%
Critical Value, z	1.645	1.960	2.576